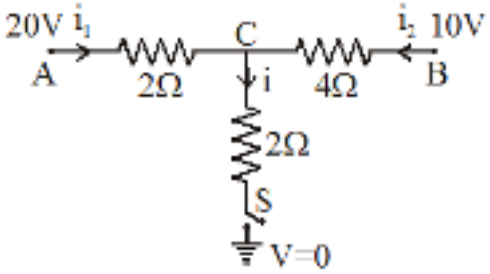
PHYSICS

1) When the switch S, in the circuit shown, is closed, then the value of current i will be :



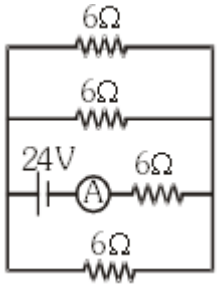
(1) 3 A

(2) 5 A

(3) 4 A

(4) 2 A

2) Reading of ammeter in the circuit below is :



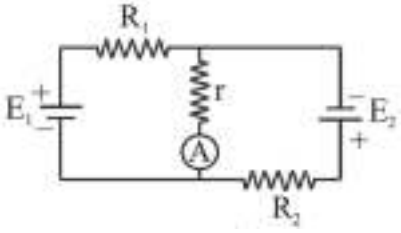
(1) 16 A

(2) 3 A

(3) 4 A

(4) 12 A

3) If the reading of ideal ammeter A in circuit shown here is zero, then the ratio $\frac{E_1}{E_2}$ of the emf's of the two batteries must be



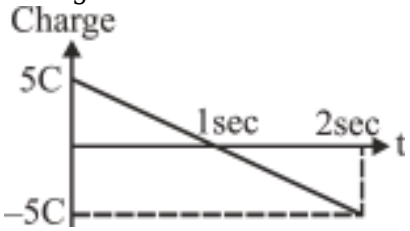
(1) 1

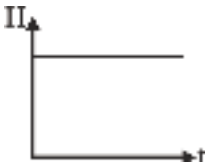
(2) $\frac{R_1}{R_2}$

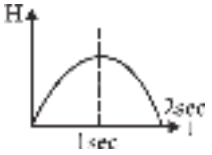
(3) $\frac{R_2}{R_1}$

(4) None of these

4) A charge passing through a resistor is varying with time as shown in the figure. The amount of heat generated in time ' t ' is best represented (as a function of time) by :



(1)


(2)


(3)


(4)


5) A galvanometer of resistance 50Ω is connected to a battery of 3V along with a resistance of 2950Ω in series. A full scale deflection of 30 divisions is obtained in the galvanometer. In order to reduce this deflection to 20 division, the resistance in series of galvanometer should be :-

(1) 6050Ω

(2) 4450Ω

(3) 5050Ω

(4) 5550Ω

6) When a metal conductor connected to left gap of a meter bridge is heated, the balancing point

(1) shifts towards right

(2) shift towards left

(3) remains unchanged

(4) remains at zero

7) Two wires A and B are made of same material, but their lengths are in the ratio of $l_1 : l_2 = 2 : 3$ and masses are in ratio of $m_1 : m_2 = 3 : 2$. The ratio of their resistances is -

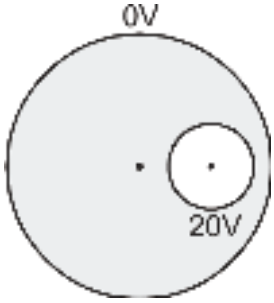
(1) 3 : 2

(2) 2 : 3

(3) 8 : 27

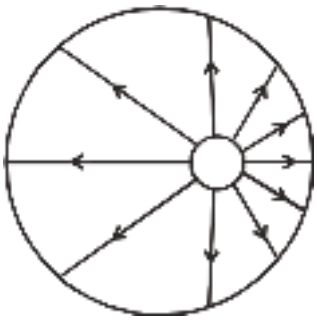
(4) 27 : 8

8) Two equipotential spherical surfaces having potential 20 V and 0 V are as shown in figure. There is no charge anywhere in space except on the surface of both the spheres. Then which of the following figure represents the nature of electric field in region between the spherical surfaces by

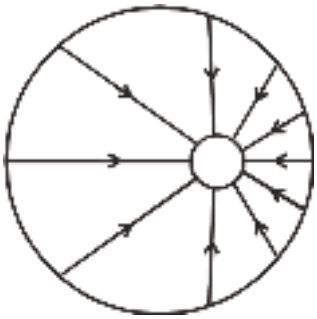


electric lines of forces.

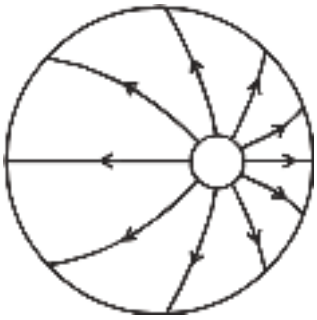
(1)



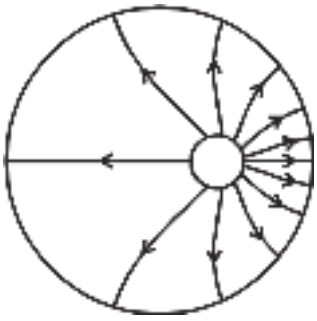
(2)



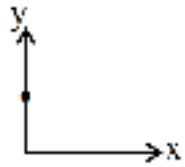
(3)

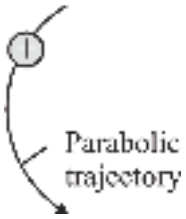


(4)



9) Which electric field is responsible for the proton's trajectory as shown below ?





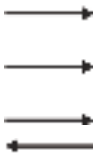
(1)



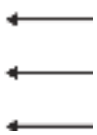
(2)



(3)



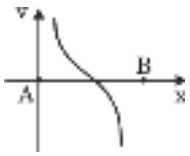
(4)



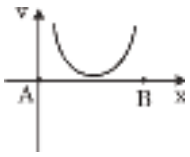
10) Two charges –q and q are on points A and B as shown. The variation of electric potential along the x-axis is :-



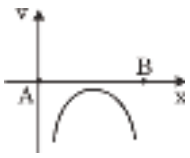
(1)



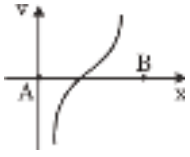
(2)



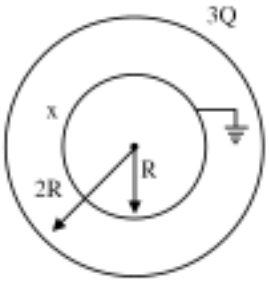
(3)



(4)



11) In the given fig. if inner sphere is grounded then charge appearing on the inner sphere.



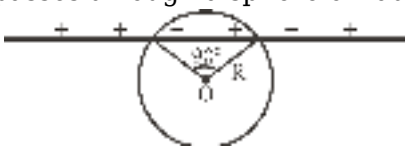
(1) 2Q

(2) $\frac{3}{2}Q$

(3) $\frac{-3}{2}Q$

(4) $\frac{2}{3}Q$

12) A uniformly charged long wire of charge density 'λ' passes through a sphere of radius 'R' as shown in figure. Find electric flux related to the sphere.



(1) $\frac{\lambda R}{\epsilon_0}$

(2) $\frac{\sqrt{2}\lambda R}{\epsilon_0}$

(3) $\frac{2\lambda R}{\epsilon_0}$

(4) Zero

13) Two positive charges of magnitude 'q' are placed at the ends of a side (side 1) of a square of side '2a'. Two negative charges of the same magnitude are kept at the other corners. Starting from rest, if a charge Q moves from the middle of side 1 to the centre of square, its kinetic energy at the centre of square is :-

(1) $\frac{1}{4\pi\epsilon_0}\frac{2qQ}{a}\left(1-\frac{1}{\sqrt{5}}\right)$

(2) zero

(3) $\frac{1}{4\pi\epsilon_0}\frac{2qQ}{a}\left(1+\frac{1}{\sqrt{5}}\right)$

(4) $\frac{1}{4\pi\epsilon_0}\frac{2qQ}{a}\left(1-\frac{2}{\sqrt{5}}\right)$

14) A particle having mass ‘m’ and charge ‘–q’ is released on the axis of a ring from a very large distance from center of the ring. The ring has radius R and charge 3q distributed uniformly on its circumference. Speed of the point charge when it reaches at a point $2\sqrt{2}R$ distance away from the center will be :

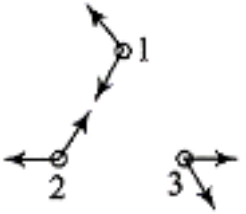
(1) $\sqrt{\frac{3q^2}{2\pi\varepsilon_0 mR}}$

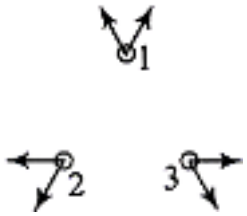
(2) $\sqrt{\frac{q^2}{6\pi\varepsilon_0 mR}}$

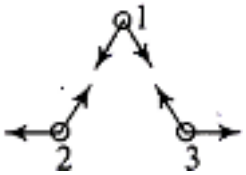
(3) $\sqrt{\frac{q^2}{2\pi\varepsilon_0 mR}}$

(4) None of these

15) Which of the following four figures correctly show the forces that three charged particles can exert on each other?

(i) 

(ii) 

(iii) 

(iv) 

(1) all of the above

(2) none of the above

(3) ii, iii

(4) ii, iii and iv

16) Capacitance of a parallel plate capacitor becomes $\frac{4}{3}$ times it's original value when a dielectric slab of thickness $\frac{d}{2}$ is inserted between the plates and is placed parallel to plates (d = the separation between the plates). The dielectric constant of the slab is

(1) 8

(2) 6

(3) 4

(4) 2

17) Two capacitors C_1 = 4μF and C_2 = 6μF are charged upto potential V_1 = 100 V and V_2 = 50 V respectively and then the sources are disconnected. plates of identical polarity of the capacitors are connected. Energy loss during the process of charge redistribution equals to

(1) 3 mJ

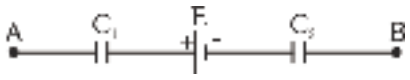
(2) 1.25 mJ

(3) 7.5 mJ

(4) 6 mJ

18)

For the section AB of a circuit shown in figure $C_1 = 1\mu F$, $C_2 = 2\mu F$, $E = 10$ V and the potential difference $V_A - V_B = -10$ volt. Then, the charge on capacitor C_1 is-



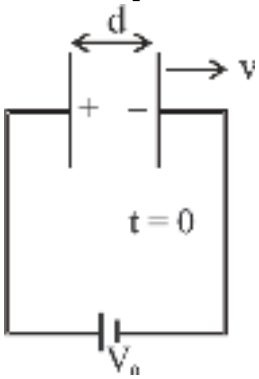
(1) 0 µC

(2) $\left(\frac{20}{3}\right)$ µC

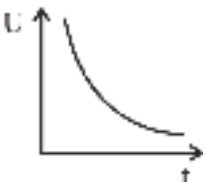
(3) $\left(\frac{40}{3}\right)$ µC

(4) None of these

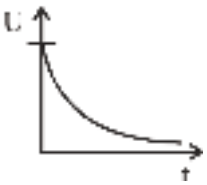
19) A parallel plate capacitor is connected to a battery and the plates are pulled apart with a uniform speed v. The graph that represents the variation in energy stored with respect to time :-



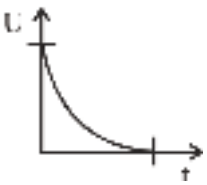
(1)



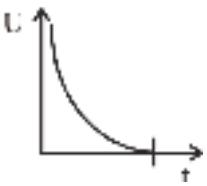
(2)



(3)



(4)



20) How many times will the mean square speed of molecules of an ideal gas become by increasing its volume to 2 times and pressure increases to 8 times and the mass remains unchanged.

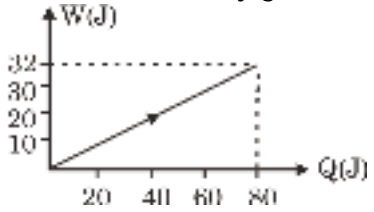
(1) 2 times

(2) 4 times

(3) 16 times

(4) Decreases by a factor of 1/4

21) In the figure shown, amount of heat supplied to one mole of an ideal gas is plotted on the horizontal axis and amount of work done by gas is drawn on vertical axis. Assuming process be isobaric i.e. gas can be :-



(1) He

(2) CO

(3) CO_2

(4) NH_3

22) A thin rod of negligible mass and area of cross-section 2×10^{-6} m^2, suspended vertically from one end, has a length of 0.5 m at 200°C. The rod is cooled to 0°C, but length is kept same by attaching a mass at the lower end. The value of this mass is : (Young's modulus = 10^{11} N/m^2, Coefficient of linear expansion 10^{-5} K^{-1}).

(1) 20 kg

(2) 30 kg

(3) 40 kg

(4) 50 kg

23) A thermometer with an arbitrary scale has the ice point at –20° and the steam point at 180°. When the thermometer reads 5°, a Centigrade thermometer will read

(1) 7.5 °C

(2) 12.5 °C

(3) 16.5 °C

(4) –9.37 °C

24) Cooling rate of a sphere having temperature 900 K and external environment temperature of 300 K is R. When the temperature of sphere is made 1200 K then new rate of cooling is -

(1) $\frac{15R}{80}$

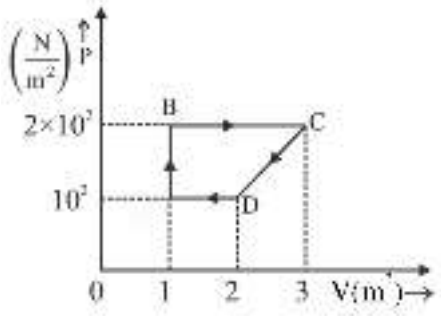
(2) $\frac{80R}{15}$

(3) $\frac{255R}{80}$

(4) $\frac{256R}{81}$

25)

A cyclic process is shown in figure. Work done during isobaric expansion is :–



(1) 150 J

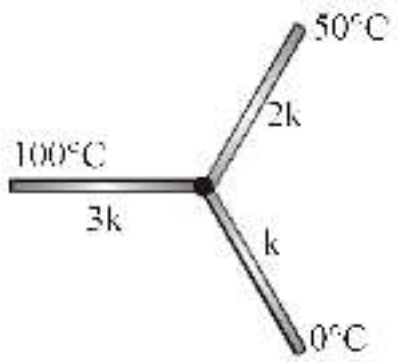
(2) 100 J

(3) 400 J

(4) 200 J

26)

Three rods of the same dimensions have thermal conductivities 3k, 2k and k. They are arranged as shown, with their ends at 100°C, 50°C and 0°C. The temperature of their junction is:–



(1) 75°C

(2) $\frac{200}{3}$°C

(3) 40°C

(4) $\frac{100}{3}$°C

27) Work done by the same sample of a gas in a process A is double the work done in another process B. The temperature rises through the same amount in both the process. If C_A and C_B are molar heat capacities for the two processes then :

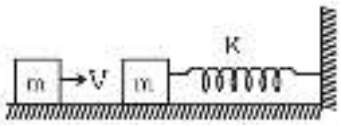
(1) $C_A = C_B$

(2) $C_A > C_B$

(3) $C_A < C_B$

(4) $C_A \leq C_B$

28) The left block collides inelastically with the right block and sticks to it. Find the amplitude of the

resulting simple harmonic motion. 

(1) $V\sqrt{\frac{m}{K}}$

(2) $V\sqrt{\frac{m}{2K}}$

(3) $2V\sqrt{\frac{m}{K}}$

(4) $V\sqrt{\frac{2m}{K}}$

29) Two simple pendulums of length 5 m and 20 m respectively are given small linear displacement in one direction at the same time. They will again be in phase when the pendulum of shorter length has completed..... oscillations.

(1) 1

(2) 4

(3) 2

(4) 3

30) A particle performs Linear SHM such that it is placed on a platform & platform along with particles oscillate vertically up and down with amplitude A = 1 cm. If the particle do not loose contact with platform anywhere and mass of particle is 1 kg then find the minimum possible time period (take $\pi = \sqrt{g}$)

(1) 0.1 sec

(2) 0.2 sec

(3) 0.3 sec

(4) 0.4 sec

31) A particle performs simple harmonic motion with amplitude A. Its speed is increased to three times at an instant when its displacement is $\frac{2A}{3}$. The new amplitude of motion is $\frac{nA}{3}$. The value of n is ___.

(1) 7

(2) 4

(3) 5

(4) 2

32) A particle of mass m moving along the x-axis has a potential energy $U(x) = a + bx^2$ where a and b are positive constants. It will execute SHM with a frequency determined by the value of

(1) b alone

(2) b and a alone

(3) b & m alone

(4) b, a and m alone

33) A particle is in simple harmonic motion along the x-axis back and forth about the origin. Its maximum acceleration is 40 cm/s^2 and amplitude is 10 cm. At the instant t = 0, it passes the position x = 5 cm moving in the positive x-direction. Which of the following equations can be used to represent this simple harmonic motion ?

(1) $x = 10\cos(2t + 30^\circ)$

(2) $x = 10\cos(2t - 30^\circ)$

(3) $x = 10\cos(2t + 60^\circ)$

(4) $x = 10\cos(2t - 60^\circ)$

34) The frequency of first overtone of a closed organ pipe of length L is f_1. A hole is made at a distance $\frac{L}{6}$ from the closed end so that it becomes an open pipe. Now the frequency of first overtone of open pipe is f_2. Then $\frac{f_1}{f_2}$ is :-

(1) $\frac{4}{5}$

(2) $\frac{5}{4}$

(3) $\frac{8}{5}$

(4) $\frac{5}{8}$

35) A uniform string of length 20 m & mass 1 kg is hung vertically. Find the speed of wave at the mid point of the string :-

(1) 20 m/s

(2) 30 m/s

(3) $10\sqrt{2}$ m/s

(4) 10 m/s

36) In a sonometer experiment, the bridges are separated by a fixed distance. The wire which may be slightly extensible, emits a tone of frequency n when held by a tension T. Tension is increased to 4T; the new tone emitted by the wire will be of frequency :-

(1) n

(2) 2n

(3) slightly greater than 2n

(4) slightly less than 2n

37) A wave travelling along the x-axis is described by the equation y (x, t) = 0.005 cos $(\alpha x - \beta t)$. If the wavelength and the time period of the wave are 0.08m and 2.0 s respectively then α and β in

appropriate units are :

(1) $\alpha = 25.00\pi,\ \beta = \pi$

(2) $\alpha = \frac{0.08}{\pi},\ \beta = \frac{2.0}{\pi}$

(3) $\alpha = \frac{0.04}{\pi},\ \beta = \frac{1.0}{\pi}$

(4) $\alpha = 12.50\pi,\ \beta = \frac{\pi}{2.0}$

38) A source of sound of frequency 600Hz is placed inside water. The speed of sound in water is 1500 m/s and in air it is 300 m/s. The frequency of sound recorded by an observer who is standing in air is:

(1) 200 Hz

(2) 3000 Hz

(3) 120 Hz

(4) 600 Hz

39) The amplitudes of the two waves are in ratio 5 : 2. If all other conditions for the two waves are same, then what is the ratio of their energy densities?

(1) 5 : 2

(2) 5 : 4

(3) 4 : 5

(4) 25 : 4

40) The first two resonance lengths formed in a resonance tube are 16.5 cm and 51 cm. The end correction for the tube is :-

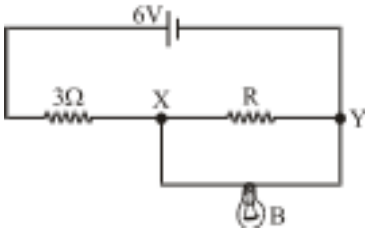
(1) 0.25 cm

(2) 0.50 cm

(3) 0.75 cm

(4) 1.00 cm

41) In the following circuit, bulb rated as 1.5 V, 0.45 W. If bulbs glows with full intensity then what



will be the equivalent resistance between X and Y.

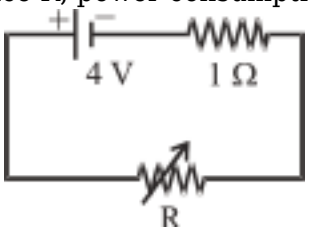
(1) 0.45 Ω

(2) 1 Ω

(3) 3 Ω

(4) 5 Ω

42) For different values of resistance R, power consumptions in R are given. Then which of the following values are not possible ?



(a) 2W (b) 5W (c) 8 W (d) 4W

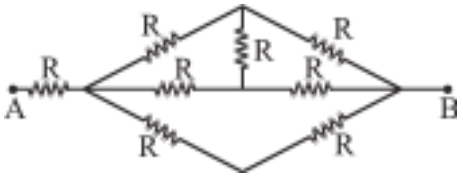
(1) Only c

(2) b & c

(3) a,b,c

(4) All

43) In the network of resistances as shown in figure. the effective resistance between points A and B is :-



(1) 8 R

(2) 5R

(3) (5/3) R

(4) (8/3) R

44) If a copper wire is stretched to make it 0.1% longer. The percentage change in its resistance is :-

(1) 0.2% increase

(2) 0.2% decrease

(3) 0.1% increase

(4) 0.1% decrease

45) For a calorimeter of water equivalent 20 gm a heated coin of copper having mass 40 gram & specific heat capacity of 420 J/kgK with temperature of 75°C is added. Equilibrium temperature comes out at 25°C. So find initial temperature of calorimeter.

(1) 7.5 °C

(2) 15 °C

(3) 10 °C

(4) 5 °C

CHEMISTRY

1) Silver is monovalent and has an atomic mass of 108. Copper is divalent and has an atomic mass of 63.3. The same electric current is passed, for the same time through a silver coulometer. If 27.0 gm of silver is deposited, then the corresponding amount of copper deposited is:-

(1) 63.60 g

(2) 31.80 g

(3) 15.90 g

(4) 7.95 g

2) Given:-
$M \rightarrow M^{2+} + 2e^-$, $E° = 0.44$ V ;
$M^{2+} \rightarrow M^{3+} + 1e^-$, $E° = -0.77$V.
Determine the value of E° for $M \rightarrow M^{3+} + 3e^-$:-

(1) –0.33 V

(2) 1.21 V

(3) +0.33 V

(4) 0.036 V

3) Calculate cell potential at 298 K for following galvanic cell
$Cd(s)\,|Cd^{+2}(aq.)\,(0.1M)\,||H^+_{(aq)}\,(0.1M)\,|H_2(g, 0.5atm)|Pt$
$E^o_{Cd^{+2}/Cd} = -0.40$ V

(1) 0.38 V

(2) – 0.38 V

(3) 0.36 V

(4) – 0.36 V

4) Which of the following will increase the voltage of the cell :-
$Sn_{(s)} + 2Ag^+_{(aq)} \rightarrow Sn^{2+}_{(aq)} + 2\,Ag_{(s)}$

(1) Increase in the concentration of Sn^{2+} ions

(2) Increase in the concentration of Ag^+ ions

(3) Increase in the size of silver rod

(4) None of these

5)

The cell reaction for the given cell is spontaneous if :
$Pt(s)|\underset{P_1}{H_2(g)}\,|\underset{(1M)}{H^+(aq)}\,||\,\underset{(1M)}{H^+(aq)}\,|\underset{P_2}{H_2(g)}\,|Pt(s)$

(1) $P_1 > P_2$

(2) $P_1 < P_2$

(3) $P_1 = P_2$

(4) $P_2 = 1$ atm

6) From the following cells, which cell does not have any gaseous species in their cell reaction :
(a) Dry cell
(b) Nickel-Cadmium cell

(c) Fuel cell
(d) Lead storage battery
(e) Mercury cell

(1) a, e

(2) a, c, d

(3) b, d, e

(4) a, b, e

7) E^{o} of some elements are given as :
$I_2 + 2e^{\ominus} \rightarrow 2I^{\ominus}$; E° = 0.54V
$MnO_4^{\ominus} + 8H^{\oplus} + 5e^{-} \rightarrow Mn^{+2} + 4H_2O; E^{o} = 1.52V$
$Fe^{+3} + e^{-} \rightarrow Fe^{+2}$; E° = 0.77 V
$Sn^{+4} + 2e^{-} \rightarrow Sn^{+2}$; E° = 0.1 V
Among these species strongest reductant is :

(1) Sn^{+2}

(2) Fe^{+2}

(3) Mn^{+2}

(4) I^{-}

8) Match the column I with column II for electrolysis of following solutions :

	Column-I		**Column-II**
(P)	$CuSO_4$ solution using Pt electrode	(A)	Mass of anode decreases
(Q)	$CuSO_4$ solution using Cu electrode	(B)	Na^{+} reduce at cathode
(R)	NaCl solution using Pt electrode	(C)	$O_{2(g)}$ evolved at anode
(S)	NaCl solution using Hg electrode	(D)	$H_{2(g)}$ evolved at cathode

(1) P-A, Q-C, R-D, S-B

(2) P-C, Q-A, R-D, S-B

(3) P-B, Q-D, R-A, S-C

(4) P-D, Q-B, R-C, S-A

9) On the basis of following data, select the correct pH value for 10^{-2} M NH_4OH(aq) solution at 25°C
$\Lambda_m(NH_4OH)$ of 10^{-2} M solution = 10.2 S-cm^2 mol^{-1}
$\Lambda_m^o(NH_4Cl)$ = 130 S-cm^2 mol^{-1}
$\Lambda_m^o(NaOH)$ = 320 S-cm^2 mol^{-1}
$\Lambda_m^o(NaCl)$ = 110 S-cm^2 mol^{-1}

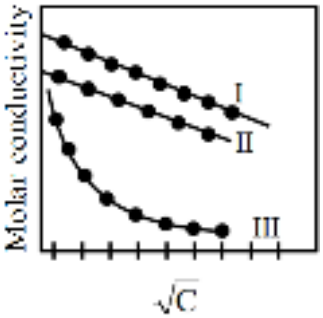
(1) 3.52

(2) 10.48

(3) 2

(4) 12

10) A graph was plotted between molar conductivity of various electrolytes (NaCl, HCl and NH_4OH)



and $\sqrt{C}$ (in mol L^{-1}). Correct set is :

(1) I(NaCl), II(HCl), III(NH_4OH)

(2) I(HCl), II(NaCl), III(NH_4OH)

(3) I(NH_4OH), II(NaCl), III(HCl)

(4) I(NH_4OH), II(HCl), III(NaCl)

11) How much charge (in Faraday) is required to produce (a) 36 g of Mg from molten $MgCl_2$ and (b) 54 g of Al from molten Al_2O_3 respectively.

(1) (a) = 2 F ; (b) = 4 F

(2) (a) = 1.5 F ; (b) = 3 F

(3) (a) = 3 F ; (b) = 6 F

(4) (a) = 2 F ; (b) = 6 F

12) The resistance of 1M CH_3COOH solution is 250 Ω , when measured in a conductivity cell with cell constant 125 m^{-1}. The molar conductivity in S cm^2 mol^{-1} is :

(1) 5×10^{-4}

(2) 500

(3) 200

(4) 5

13) Suitable indicator for the acid-base titration of HCOOH against KOH is.

(1) Methyl orange

(2) Methyl red

(3) Phenolphthalein

(4) Thymol Blue

14) What volume of 0.1 M $KMnO_4$ solution is required to oxidise 50 mL, 0.1 M oxalic acid in acidic medium

(1) 10 mL

(2) 20 mL

(3) 30 mL

(4) 50 mL

15) Which of the following statements are correct :
(i) When freshly prepared precipitate are converted into colloidal solution then process is called peptization.
(ii) Lyophilic sol are mostly of inorganic nature.
(iii) Lyophilic sol are highly solvated as particles have great affinity for solvent.
(iv) Lyophobic sols are reversible sols so special methods are required to prepare them.

(1) Only i
(2) i and iii
(3) ii and iv
(4) All of these

16) Henry's law constant for dissolution of CH_4 in benzene at 298 K is 2×10^5 mm of Hg. Then solubility of CH_4 in benzene at 298 K (in terms of mole fraction) under 760 mm of Hg is :

(1) 1.2×10^{-5}
(2) 3.8×10^{-3}
(3) 4×10^{-7}
(4) 1×10^{-2}

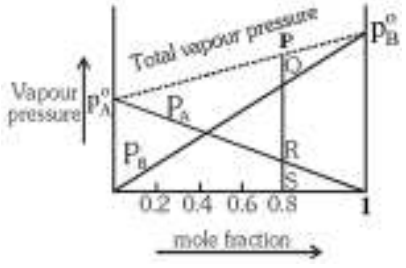
17) Which solution will have the highest boiling point :-

(1) 1% w/v glucose in water
(2) 1% w/v sucrose in water
(3) 1% w/v NaCl in water
(4) 1% w/v $CaCl_2$ in water

18) The vapour pressure of a solution of a non-volatile non-electrolyte (A) in a solvent (B) is 95 % of the vapour pressure of the solvent at the same temperature. If $M_B = 0.3\ M_A$, where M_B and M_A are molecular weight of B and A respectively, the weight ratio of the solvent and solute are :-

(1) –0.17
(2) 5.7
(3) 0.2
(4) 4.0

19) Consider the following vapour pressure-composition graph, SP is equal to :-



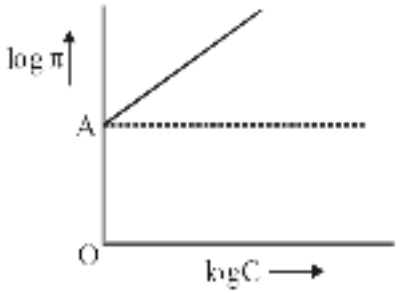
(1) PQ + RS
(2) QR + RS

(3) SR + SQ

(4) PQ + QR

20)

Variation of logπ (atm) v/s logC
(C = concentration of solute in g/L) is given



At 300K, for a solute of molar mass 300 g/mol, OA = log ($x \times 10^{-2}$).
What is the value of x ?

(1) 8.21

(2) 0.0821

(3) 82.1

(4) 0.821

21) **Statement 1 :** In endothermic reaction threshold energy for forward reaction is more than backward reaction.
Statement 2 : As the temperature increases, the number of reactant molecules which can cross the potential energy barrier increases sharply.

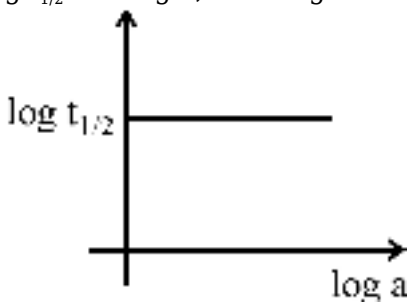
(1) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(2) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(3) Statement-1 is true, statement-2 is false.

(4) Statement-1 is false, statement-2 is true.

22) A graph between log $t_{1/2}$ and log a, 'a' being the initial concentration of reactant A is given, for reaction : A → product.



If 75% of A is decomposed in 100 min. How long will it take for 90% of the compound to decompose.
[Given : log 2 = 0.3]

(1) 176.66 min.

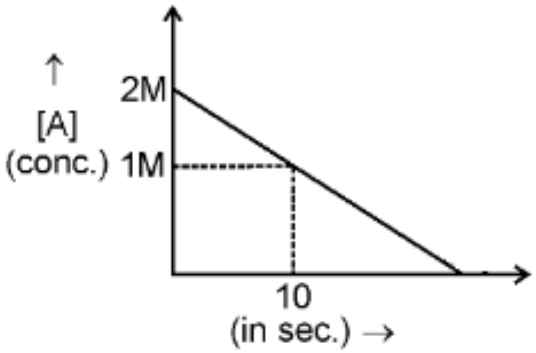
(2) 156.66 min.

(3) 166.66 min.

(4) 150 min.

23) For a zero order reaction, $A \xrightarrow{k} P$

Variation of concentration of A with time is given as:



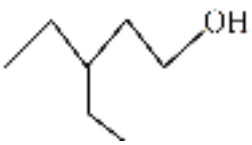
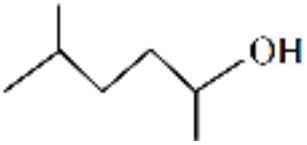
Then calculate time to reduce concentration of A from 2 M to 0.5 M

(1) 10 sec

(2) 20 sec

(3) 15 sec

(4) 25 sec

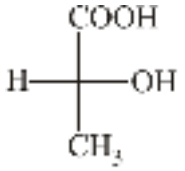
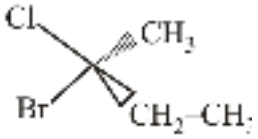
24)  and  are -

(1) Position isomer

(2) Chain isomer

(3) Position and chain both isomer

(4) Identical

25)  (I)  (II)

The correct configuration assigned for compound I & II respectively are :-

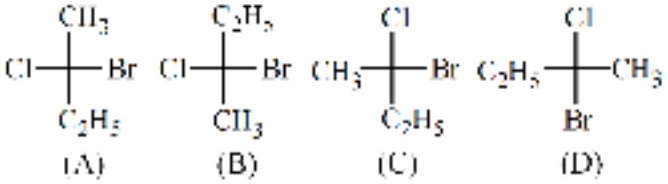
(1) R, R

(2) S, S

(3) R, S

(4) S, R

26) Consider the following structures (A), (B), (C) and (D)



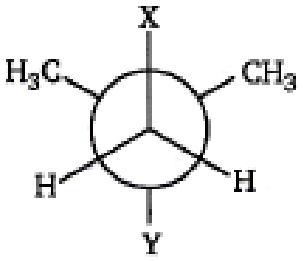
Which of the following statements is not correct?

(1) B and C are identical

(2) A and B are enantiomers

(3) A and C are enantiomers

(4) B and D are enantiomers

27) In the Newman projection for 2, 2-dimethylbutane 

X and Y are respectively :

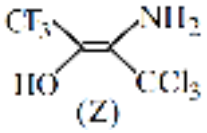
(1) H and H

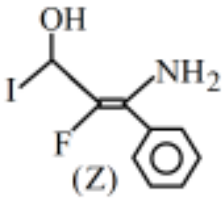
(2) H and C_2H_5

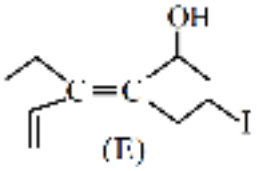
(3) C_2H_5 and H

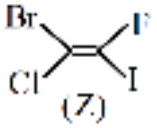
(4) propane

28) Identify the compound having correct E–Z notation.

(1) 

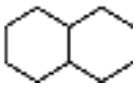
(2) 

(3) 

(4) 

29) Identify correct statement :

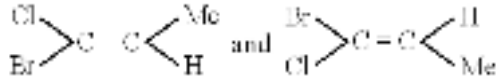
(1) Compound having D-configuration is always dextrorotatory in nature.

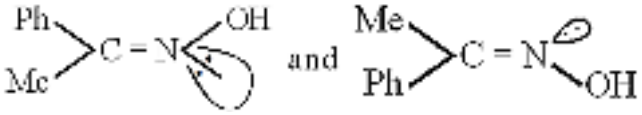
(2)  does not have any chiral centre

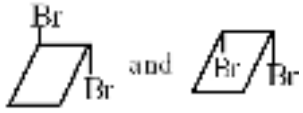
(3) Geometrical isomers have same boiling point

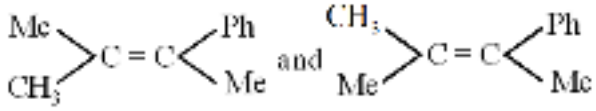
(4) Enantiomers have different numerical value of angle of rotation

30) Which is a pair of geometrical isomers

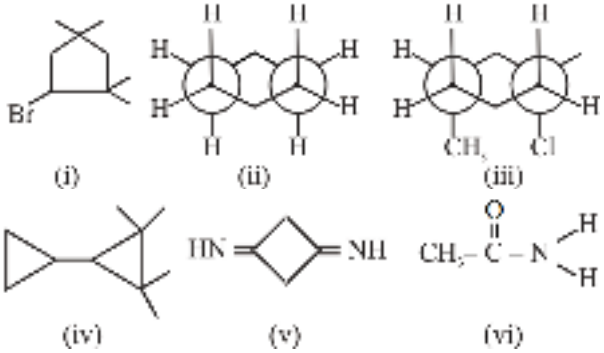
(1)  and

(2) 

(3) 

(4) 

31) How many of following compound will show G.I :-



(1) 4

(2) 5

(3) 2

(4) 6

32) A mixture of o-nitrophenol and p-nitrophenol can be separated by :-

(1) Simple distillation

(2) Fractional distillation

(3) Vacuum Distillation

(4) Steam Distillation

33) Match the methods of separation (I) with the given mixture (II).

	Method (I)		**Mixture (II)**
(A)	Crystallisation	(P)	sugar + water
(B)	Sublimation	(Q)	sugar + naphthalene
(C)	Distillation	(R)	sugar + charcoal
(D)	Chromatography	(S)	chloroform + aniline
		(T)	glycine + alanine

	A	**B**	**C**	**D**

(1)	(P,R)	(Q)	S	T
(2)	(P,Q)	(R,S)	T	P
(3)	(P,R)	(Q,S)	P	T
(4)	R	(P,Q)	T	S

(1) 1

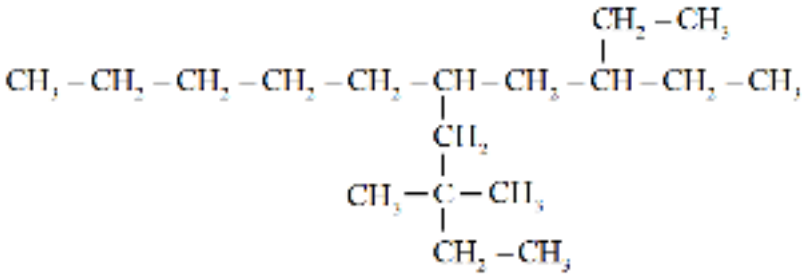
(2) 2

(3) 3

(4) 4

34)

IUPAC name of



(1) 5-(2,2-Dimethylbutyl)-3-ethyl decane

(2) 5-(2-Ethylbutyl)-3,3-dimethyl decane

(3) 7-Ethyl-3,3-dimethyl-5-pentylnonane

(4) 3-Ethyl-7,7-dimethyl-5-pentylnonane

35) In carius method of estimation of halogen 0.15g of an organic compound gave 0.12 g of AgBr. Find out the percentage of bromine in the compound :-

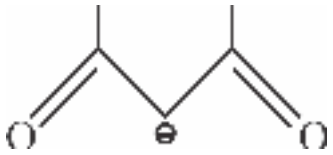
(1) 34.04%

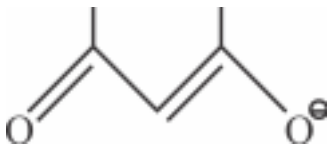
(2) 58.20%

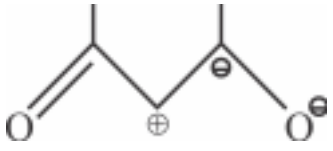
(3) 9.24 %

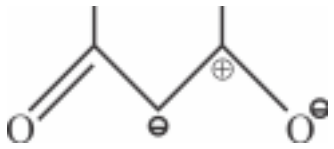
(4) 0.12 %

36) Which of the following is most stable resonating structure?

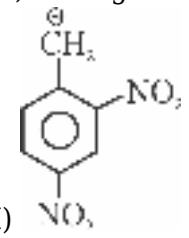
(1) 

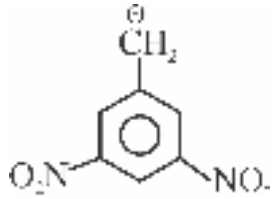
(2) 

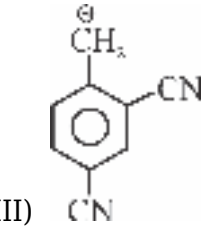
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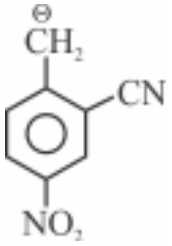
(4) 

37) Arrange the following carbanions in decreasing order of stability:

(I) 

(II) 

(III) 

(IV) 

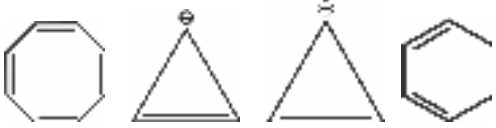
(1) III > IV > I > II

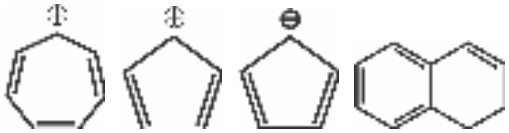
(2) I > III > IV > II

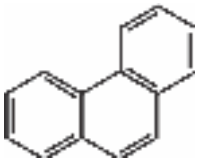
(3) I > II > III > IV

(4) I > IV > III > II

38) Among the following, the number of aromatic compound(s) is







(1) 4

(2) 5

(3) 6

(4) 3

39) How many statement is correct in following

(a) The difference in energy between the actual structure and the resonating structure is called resonance energy

(b) Resonance structure have same position of nuclei and same number of unpaired electron

(c) Among the resonance structure the one which has more number of covalent bond is most stable

(d) Among the resonance structure negative charge at more electronegative atom and positive

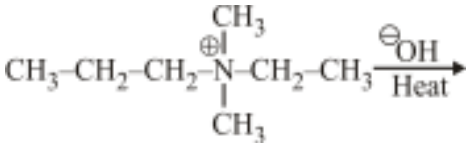
charge at less electronegative atom is more stable
(e) Complete octet molecule is more stable than incomplete octet molecule.

(1) 3

(2) 4

(3) 5

(4) 2



40)
Major product will be ?

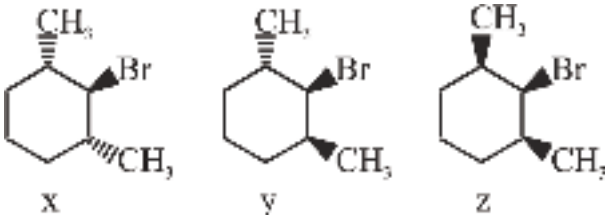
(1) CH_3–CH=CH_2

(2) CH_3–CH_2–CH_3

(3) CH_2=CH_2

(4) CH_3–CH_3

41) Arrange the following in decreasing order of E_2 reaction :-



(1) x > y > z

(2) x > z > y

(3) y > z > x

(4) z > y > x

42)



Me–C≡C–Me →
Identify best reagent sequence for above conversion ?

(1) (i) Ni / H_2 ; (ii) Br_2 / CCl_4

(2) (i) Na / Liq. NH_3 ; (ii) Br_2 / CCl_4

(3) Br_2 / CCl_4 (excess)

(4) HBr / CCl_4 (excess)

43) Identify the set of reagents/reaction conditions 'X' and 'Y' in the following set of transformations :-

$CH_3–CH_2–CH_2Br \xrightarrow{X}$ Product $\xrightarrow{Y}$ $CH_3-\underset{\underset{Br}{|}}{CH}-CH_3$

(1) X = dilute aqueous NaOH, 20°C,
Y= Br_2/acetic acid, 20°C

(2) X = concentrated alcoholic NaOH, 80°C;
Y = HBr/$CHCl_3$, 20°C

(3) X = dilute aqueous NaOH 20°,
Y = Br_2/$CHCl_3$, 0°C

(4) X = concentrated alcoholic NaOH, 80°C;
Y = Br_2/$CHCl_3$, 0°C

44) ***Assertion (A) :-*** Chlorobenzene is less reactive than benzene for electrophilic substitution reaction.
Reason (R) :- Resonance effect in chlorobenzene destabilized carbocation intermediate, so it is m-directing.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**

45) Which of following would not give 2-phenyl butane as the major product in a Friedel Crafts alkylation reaction with benzene?

(1) Butanoyl chloride + $AlCl_3$ then Zn–Hg/HCl

(2) n-Butyl chloride + $AlCl_3$

(3) 2-Butanol + H_2SO_4

(4) 1-Butene + HF/0°C

BIOLOGY

1)

Identify the correct match from columns I, II and III :-

	Column-I		**Column-II**		**Column-III**
1	Chlorophyll 'a'	a	Absorption peak at 680 nm	i	Prevents Photo-oxidation
2	Photosystem-I	b	Accessory pigments	ii	Only grana Lamellae
3	Photosystem-II	c	Chief pigment	iii	Blue green in colour

4	Carotenoids	d	Absorption peak at 700 nm	iv	Present in both grana and stroma lamellae

(1) 1 - c - iii, 2 - d - iv, 3 - a - ii, 4 - b - i

(2) 1 - b - i, 2 - a - ii, 3 - c - iii, 4 - d - iv

(3) 1 - b - iv, 2 - c - iii, 3 - d - ii, 4 - a - i

(4) 1 - b - i, 2 - a - ii, 3 - d - iii, 4 - c - iv

2) Read the following four statements (A - D) :-

(A) There is a linear relationship between light and CO_2 fixation rate at low light intensity but at higher light intensity, it is not

(B) Current availability of CO_2 is limiting to the C_3 plants because they show CO_2 saturation at 360 $\mu l L^{-1}$ concentration.

(C) Light reactions of photosynthesis are less affected by temperature as compared to dark reactions

(D) As a factor water mainly show indirect effect on photosynthesis

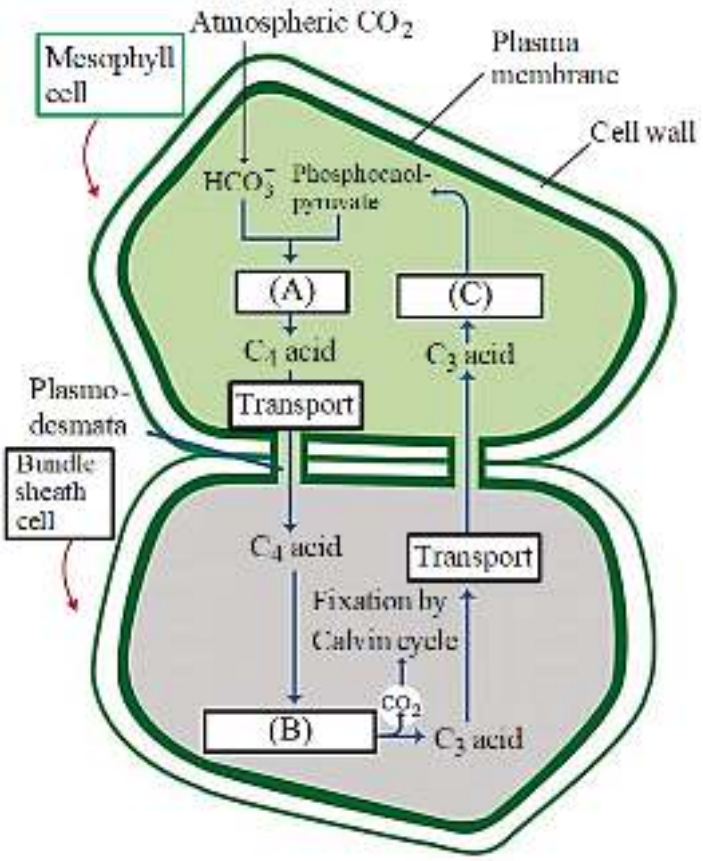
Which of the above statement(s) is/are **right** ?

(1) A, C, D

(2) Only B

(3) B and C

(4) Only A



3) Identify the **A**, **B** and **C** ?

(1) **A**-Carboxylation, **B**-Reduction, **C**-Regeneration

(2) **A**-Carboxylation, **B**-Regeneration, **C**-Reduction

(3) **A**-Carboxylation, **B**-Decarboxylation, **C**-Regeneration

(4) **A**-Fixation, **B**-Carboxylation, **C**-Reduction

4) **Assertion (A) :** During light reaction, protons are accumulated within lumen of thylakoid.
Reason (R) : During light reaction, splitting of water molecules takes place on inner side of thylakoid membrane

(1) Both (A) and (R) are true, and (R) is correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

5) Which of the following is **incorrect** about the light reactions of photosynthesis ?

(1) P_{680} and P_{700} are the reaction centres of PS-I, and PS-II respectively

(2) Light energy provides energy for the photolysis of water through excitation of the reaction centre of PS-II

(3) NADPH is not produced in cyclic electrons transport in light reactions

(4) Interaction of both the photosystems is needed for the reduction of $NADP^+$

6) In photosynthesis, the electrons that are moved from photosystem- II must be replaced. This is achieved by electrons available :-

(1) From the photosystem-I

(2) From the $NADPH_2$

(3) From the splitting of water

(4) From the cytochrome

7) During dark reaction of photosynthesis which of the following event takes place ?

(1) Molecular oxygen is released

(2) PGAL is synthesized

(3) ATP is formed

(4) Hydrogen is released

8) By the production of which of the following it can be identified that whether CO_2 or O_2 bind to the active site of RuBisCO ?

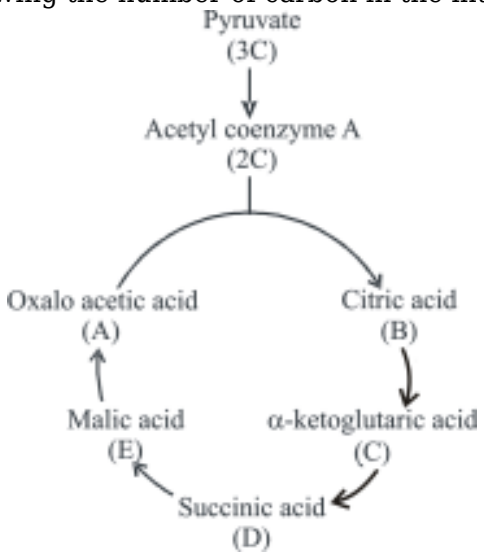
(1) Phosphoglycolate

(2) NADPH

(3) ATP

(4) Phosphoglycerate

9) Choose the correct option of labeling showing the number of carbon in the intermediate compounds, involved in the citric acid cycle.



(1) (A)-4C, (B)-6C, (C)-5C, (D)-4C, (E)-4C

(2) (A)-6C, (B)-5C, (C)-4C, (D)-3C, (E)-2C

(3) (A)-2C, (B)-3C, (C)-4C, (D)-5C, (E)-6C

(4) (A)-4C, (B)-6C, (C)-4C, (D)-4C, (E)-4C

10) **Statement-I :-** Two successive steps of decarboxylation in TCA cycle lead to the formation of α ketoglutaric acid.
Statement-II :- The TCA cycle starts with the condensation of acetyl group with oxaloacetic acid and water.

(1) Statement I and Statement II both are incorrect.

(2) Statement I and Statement II both are correct.

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

11) Match the **column-I** with **column-II** and choose the option with correct match :-

	Column-I [Names of complexes of mitochondrial ETS]		**Column-II [Member or complex]**
(a)	Succinate dehydrogenase	(1)	Cytochrome a and a_3
(b)	Cytochrome 'C' oxidase	(2)	FAD
(c)	NADH dehydrogenase	(3)	Complex - III
(d)	Cytochrome bc_1	(4)	FMN

Options :-

(1) a-1, b-2, c-4, d-3

(2) a-4, b-1, c-2, d-3

(3) a-2, b-1, c-4, d-3

(4) a-2, b-4, c-1, d-3

12)

Which of the following is **not correct** regarding fermentation?

(1) It accounts for only a partial breakdown of glucose

(2) It generates a net gain of only 2 molecules of ATP for each molecule of glucose

(3) NADH is oxidised to NAD^+ rather rapidly as compared to aerobic respiration

(4) It occurs in anaerobic conditions

13) How many ATP are produced in glycolysis by substrate level phosphorylation, if 3 molecules of glucose enter in respiration ?

(1) 4

(2) 8

(3) 12

(4) 10

14)

RQ value of tripalmitin is :-

(1) $\frac{145CO_2}{102O_2}$

(2) $\frac{102O_2}{145CO_2}$

(3) $\frac{102CO_2}{145O_2}$

(4) $\frac{6O_2}{6CO_2}$

15) The last electron acceptor of respiration in an aerobic organism is :-

(1) Cytochrome

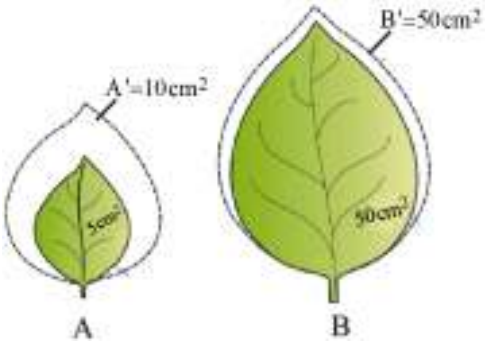
(2) Oxygen

(3) Hydrogen

(4) Ferredoxin

16)

Two leaves (A & B) with their growth are depicted below consider A' and B' are final size after growth.



Which of the following statements explain the above situation **correctly** ?

(a) Absolute growth is same in A and B
(b) Relative growth of A is twice as that of B
(c) Relative growth of B is same as relative growth of A
(d) Relative Growth of B is 1/10 of relative growth of A

Options :-

(1) b, c

(2) a, d

(3) a, c

(4) b, d

17) Ethylene, GA_3, IAA, IBA, kinetin, ABA
How many of the above are largely growth promotor hormones.

(1) 3

(2) 2

(3) 6

(4) 4

18) Select **correct** functions of gibberellin -
(a) Speed up the malting process
(b) Spraying juvenile conifers with GAs hastens the maturity
(c) Promote bolting in beet, cabbages and other plants with rosette habit.

(1) Only a & c

(2) Only b & c

(3) Only a & b

(4) a, b & c

19)

Match the columns :

	Column-I		**Column-II**
A	Cytokinin	I	Respiratory Climactic
B	Abscisic Acid	II	Anti-gibberellins
C	Ethylene	III	Coconut milk

D	Auxin	IV	First isolated from human urine

(1) A-IV, B-III, C-I, D-II

(2) A-IV, B-I, C-II, D-III

(3) A-III, B-II, C-I, D-IV

(4) A-III, B-I, C-II, D-IV

20) Which of the following conclusion was proved by using radioisotopic technique with respect to photosynthesis ?

(1) The green part of the plants is not essential

(2) CO_2 is not required

(3) O_2 evolved by the green plant comes from H_2O, not from CO_2

(4) Light is not necessary

21) Which hormone helps in lateral shoot growth ?

(1) Ethylene

(2) Cytokinin

(3) Auxin

(4) GA

22) The maximum growth rate occurs in :-

(1) Exponential phase

(2) Stationary phase

(3) Senescent phase

(4) Lag phase

23) Heterophylly is seen in :

(1) *Ranunculus* (Butter cup)

(2) Coriander

(3) Larkspur

(4) All the above

24)

In the list given below how many members have axile placentation ?

Petunia, Ladyfinger, Sunhemp, China rose, Dianthus, Lemon, Lilium, Cauliflower, Primrose, Orange, Chilly, Tomato, Tulip

(1) Six

(2) Seven

(3) Eight

(4) Nine

25)

Consider the following statements (A–D) and select the option which includes all the **correct** ones :

(A) A leaf is said to be simple when its lamina is entire or when incised, the incision do not touch the mid rib or petiole. (B) A compound leaf has axillary bud in its axil. (C) Alternate phyllotaxy found in *Calotropis* and Guava. (D) Reticulate venation is found in mango.

(1) A, B, D

(2) Only A

(3) Only A and B

(4) Only A and C

26) **Statement-I :-** In the flowers of brinjal and tulip the stamens are epipetalous and epiphyllous, respectively.
Statement-II :- In caster and pea plants, the seeds are endospermic and non-endospermic respectively.
Choose the correct option :-

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) Only **Statement I** is correct

(4) Only **Statement II** is correct

27)

Find out the correct match from the following:-

	Column-I	**Column-II**	**Column-III**
(1)		Valvate	Cotton
(2)		Twisted	Chinarose
(3)		Imbricate	*Cassia*
(4)		Vexillary	Bean

(1) 1

(2) 2

(3) 3

(4) 4

28) Development of flowers in a cymose inflorescence is:-

(1) Centripetal
(2) Bidirectional
(3) Basipetal
(4) Acropetal

29) Inferior ovary is present in :-

(1) Hypogynous flower
(2) Perigynous flowers
(3) dichogamous flower
(4) Epigynous flower

30) The correct floral formula of *Atropa belladona* is :-

(1) $Br \oplus ♂♀ P_{3\ 3} A_{3\ 3} \underline{G_{(3)}}$

(2) $\oplus ♂♀ K_{(5)} \overset{\frown}{C_{(5)}} A_5 \underline{G_{(?)}}$

(3) $\% ♂♀ K_{(5)} C_{1+2-(2)} A_{(9)+1} G_1$

(4) $♂♀ K_{2+2} C_4 A_{2+4} \underline{G_{(2)}}$

31) Choose the correct sequence of the layers from outer side to inner side in a woody dicot stem :

(1) Cork, Cork cambium, Secondary cortex, Primary phloem, Secondary Phloem, Vascular cambium, Secondary xylem, Primary xylem
(2) Cork, Cork cambium, Secondary Xylem, Secondary Phloem, Primary Phloem, Secondary cortex, Vascular cambium, Primary Xylem
(3) Primary Xylem, Secondary xylem, Vascular cambium, Primary phloem, Secondary Phloem, Secondary cortex, Cork cambium, Cork
(4) Primary phloem, Secondary Phloem, Secondary Xylem, Primary xylem, Vascular cambium, Secondary cortex, Cork cambium, Cork

32)

Read the following five statements (A-E).
A.The collenchyma occurs in layers below the epidermis in monocotyledonous plant.
B. Xylem parenchyma cells are living and thin walled and their cell walls are made up of lignin.
C. Phloem fibres are generally present in the primary phloem.
D. Tracheids are the main water transporting elements in gymnosperm.
E. Sieve tube elements are devoid of nucleus at maturity.

How many of the above statements are correct ?
(1) Four
(2) One
(3) Two

(4) Three

33) **Assertion (A) :** Arrangement of vascular bundles in roots is called radial.
Reason (R) : In radial type of vascular bundles xylem and phloem are arranged in alternate manner along different radii.
Select the correct option.

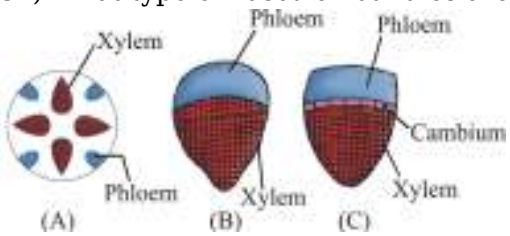
(1) (A) is true but (R) is false.

(2) Both (A) and (R) are false.

(3) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(4) Both (A) and (R) are true and (R) is the correct explanation of (A).

34) What type of vascular bundles are A, B and C respectively ?



(1) Radial; Close collateral conjoint; Open collateral conjoint

(2) Close collateral conjoint,; open collateral conjoint; radial

(3) Open collateral conjoint; Close collateral conjoint; radial

(4) Bicollateral; concentric; radial

35) Mesophyll is differentiated into palisade & spongy tissue in :-

(1) Both monocot and dicot leaves

(2) Submerged hydrophytic leaves

(3) Monocot leaves

(4) Dicot Leaves

36) Which of the following is part of ground tissue system in monocot stem :-

(1) Hypodermis

(2) Pericycle

(3) Epidermis

(4) Both (1) & (3)

37) Read the following five statements (A-E) and answer as asked next to them:
(A) Pollination by water is more common amongst abiotic pollinations.
(B) Pollination by water is quite rare in flowering plants.
(C) In water hyacinth and water lily, the flowers emerge above the level of water and are pollinated by water.
(D) In wind pollinated plants numerous flowers are packed into an inflorescene.
(E) Self incompatibility is an out breeding device which prevents inbreeding. How many of the above statements are **correct** ?

(1) Four

(2) One

(3) Two

(4) Three

38) In angiosperms following events occur during microgametogenesis :-
(a) Mitotic division in microspore.
(b) Meiotic division in miorospore mother cell.
(c) Formation of pollen grain.
(d) Formation of microspore.
(e) Formation of mature male gametophyte.
The correct sequence of event is :-

(1) b, a, d, e, c

(2) c, b, a, e, d

(3) b, d, a, c, e

(4) d, e, b, a, c

39) All of the following include out breeding devices except:

(1) Bisexuality (Monocliny), Homogamy, Cleistogamy.

(2) Anthers and stigma are placed at different position.

(3) Pollen release and stigma receptivity are not synchronised.

(4) Unisexuality of flowers and self incompatibility.

40) Which of the following is a correct match :

	Column-I	**Column-II**
(1)	♂ gamete + ♀ gamete	Endosperm
(2)	♂ gamete + Polar nuclei	Syngamy
(3)	♂ gamete + Secondary nucleus	Triple fusion
(4)	♂ gamete + ♀ gamete	Synergid cell

(1) 1

(2) 2

(3) 3

(4) 4

41) **Assertion (A) :** The megaspore mother cell undergoes meiotic division to form megaspores in the nucellus.
Reason (R) : Ovules generally differentiate a single megaspore mother cell in the micropylar region of the nucellus to form haploid megaspore.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is not correct

(4) (A) is not correct but (R) is correct

42) **Statement-I** : During embryo sac formation megaspore divide by free nuclear division.
Statement-II : In these mitotic divisions karyokinesis is not immediately followed by cytokinesis.

(1) Statement-I & Statement-II are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II is incorrect.

43) Coleoptile and coleorrhiza are the protective sheath covering of_____ and _____ respectively :-

(1) Plumule, epicotyl

(2) Radicle, plumule

(3) Plumule, radicle

(4) Radicle, hypocotyl

44) An ancient seed that is excavated from Arctic Tundra, germinated & flowered after an estimated record of 10,000 years of dormancy is:-

(1) Tulip

(2) *Phoenix*

(3) *Petunia*

(4) *Lupinus*

45) If hybrids are made into apomicts, then :-

(1) Characters may be segregate in the hybrid progeny.

(2) Hybrid will show inferior characters only.

(3) There is no segregation of characters in the hybrid progeny.

(4) Hybrid seeds will not develop into hybrid plants.

46) Read following terminology of column-I and column-II and choose correct option ?

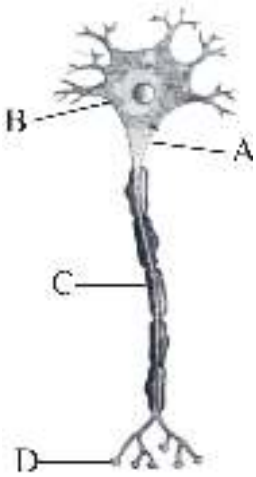
Column-I		**Column-II**	
(A)	Strong uterine contraction	(i)	Prolactin
(B)	Release of ovum	(ii)	Oxytocin
(C)	Humoral immunity	(iii)	Leutenising hormone
(D)	Stimulate reabsorption of water	(iv)	Thymus
		(v)	Vasopressin
		(vi)	Estrogen

(1) (A)-(i), (B)-(iii), (C)-(iv), (D)-(ii)

(2) (A)-(i), (B)-(ii), (C)-(iv), (D)-(vi)

(3) (A)-(ii), (B)-(iii), (C)-(iv), (D)-(v)

(4) (A)-(ii), (B)-(iv), (C)-(vi), (D)-(iii)



47) Given below is a diagrammatic representation or structure of a neuron :-
Choose the correct option indicating the labelled structures in which Nissl's granules are present :-

(1) Only B

(2) Only A, B and C

(3) Only A, B and D

(4) A, B, C and D

48) Which statement is false?

(1) Neurons are excitable cells because their membrane is in polarised state

(2) Myelinated nerve fibres are not found in spinal and cranial nerves

(3) Neurons can detect, receive and transmit different kinds of stimuli

(4) Unipolar neurons are usually found in embryonic stage

49) Choose the correct nature/characteristic of hormone :-
(a) Non-nutrient chemicals
(b) Intercellular messenger
(c) Product in trace amount

(1) a and b

(2) b and c

(3) a and c

(4) a, b and c

50) Both the lobes of thyroid gland is connected by ?

(1) Thick flap of epithelial tissue

(2) Thin flap of epithelial tissue

(3) Thick flap of connective tissue

(4) Thin flap of connective tissue.

51) Parathyroid hormone ?

(1) Acts on bones and stimulate bone resorption.
(2) Inhibits the reabsorption of Ca^{++} by renal tubules.
(3) Inhibits Ca++ ions absorption from digested food.
(4) All of the above

52) In process of spermatogenesis which hormone play a major stimulatory role ?

(1) Estrogen
(2) Androgen
(3) Progesterone
(4) Relaxin

53) **Statement-I** : The number of cervical vertebrae are seven in almost all mammals including human beings.
Statement-II : Lumber vertebrae, ribs and sternum together form the rib cage.

(1) Both **Statement I** and **Statement II** are true.
(2) Both **Statement I** and **Statement II** are false.
(3) **Statement I** is true but **Statement II** is false.
(4) **Statement I** is false but **Statement II** is true.

54) **Statement I :** Muscle fibre is the functional unit of muscle.
Statement II : Each muscle fibre has many parallelly arranged myofibrils.

(1) **Statement I** is true and **Statement II** is false
(2) **Statement I** is false and **Statement II** is true
(3) Both **Statement I** and **Statement II** are true
(4) Both **Statement I** and **Statement II** are false

55) Which of the following is not present in globular head of myosin ?

(1) An active ATPase enzyme.
(2) Active sites for actin.
(3) Binding sites for ATP.
(4) Active sites for calcium.

56) **Assertion :** Cyclic menstruation is an indicator of normal reproductive phase and extends between menarche and menopause.
Reason : During pregnancy all events of the menstrual cycle stop and there is no menstruation.

(1) A is correct but R is not correct.
(2) A is not correct but R is correct.
(3) Both A and R are correct and R is the correct explanation of A.
(4) Both A and R are correct and R is not the correct explanation of A.

57) Read following events carefully and choose correct sequence of these events for human :-
(a) Gestation (b) Gametogenesis
(c) Fertilization (d) Implantation
(e) Insemination

(1) b → e → d → c → a
(2) b → c → e → d → a
(3) b → e → c → d → a
(4) c → e → b → a → d

58) First movement of foetus & appearance of hair are observed during which month ?

(1) Second month
(2) Seventh month
(3) Fifth month
(4) Ninth month

59) Foetal ejection reflex triggers release of :-

(1) Oxytocin from the maternal ovary
(2) Oxytocin from the placenta
(3) Oxytocin from the maternal pituitary
(4) Oxytocin from the foetal umbilical cord.

60) **Statement-I :** Male accessory ducts help in storage and transfer of spermatozoa.
Statement-II : Male accessory ducts are secondary reproductive organs.

(1) Both statement-I and statement-II are incorrect
(2) Statement-I is correct but statement-II is incorrect
(3) Statement-I is incorrect but statement-II is correct
(4) Both statement-I and statement-II are correct

61) Match list-I with list-II relating to examples of various kind of contraceptive methods :-

	List-I		**List-II**
(A)	Natural method	(I)	Condoms
(B)	Barrier method	(II)	Coitus interruptus
(C)	Multiload 375	(III)	Hormone releasing
(D)	LNG-20	(IV)	Copper releasing

Choose the correct answer from the option given below :-
(1) A-II, B-I, C-IV, D-III
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-I, C-III, D-IV

(4) A-III, B-II, C-IV, D-I

62) Match list-I with list-II relating to assisted reproductive technologies.

	List-I		**List-II**
(A)	Transfer to ovum into fallopian tube	(I)	ZIFT
(B)	Transfer of early embryos with upto 8 blastomere	(II)	GIFT
(C)	Transfer of embryo more than 8 blastomeres	(III)	ICSI
(D)	A sperm is directly injected into the ovum	(IV)	IUT

(1) A-II, B-I, C-IV, D-III

(2) A-IV, B-I, C-II, D-III

(3) A-I, B-II, C-IV, D-III

(4) A-II, B-I, C-III, D-IV

63) Medical termination of pregnancy is considered relatively safe upto _____ of pregnancy.

(1) 12 weeks

(2) 15 weeks

(3) 20 weeks

(4) 24 weeks

64) Choose the correct statement :-

(1) The primary function of ganglia is to generate neurotransmitters that stimulate post-synaptic neurons at synapses.

(2) Both sympathetic and parasympathetic neurons release Ach.

(3) Ach is the main neurotransmitter that stimulates muscles at NMJ

(4) Neurotransmitter levels of sympathetic nervous system are generally stable and do not fluctuate for different situation.

65) The neural system of animals is composed of neurons, which one is correct for neuron ?

(1) Only detect stimuli

(2) Only detect and receive stimuli

(3) Detect, receive and transmit same kind of stimuli

(4) Detect, receive and transmit different kind of stimuli

66) What happens when neurotransmitter binds to receptors on post synaptic membrane ?

(1) Vesicles are released.

(2) New potential is developed

(3) Ion-channels close

(4) Axon terminals fuse.

67) Select the incorrect statement :-

(1) The pituitary gland is located in a bony cavity called sella tursica and is attached to hypothalamus by a stalk.

(2) Pituitary is divided anatomically into a adenohypophysis and a neurophypophysis.

(3) Pars distalis secretes only one hormone called melanocyte stimulating hormone.

(4) In humans, the pars intermedia is almost merged with pars distalis.

68) Match the List-I with List-II

List-I		**List-II**	
A	Para thyroid hormone	I	blood calcium level
B	Epinephrine and nor epinephrin	II	Adrenal medulla
C	Thyroid stimulating hormone	III	Anterior pituitary
D	Melatonin	IV	Sleep wake cycle

Choose the correct answer from the options given below

(1) A-III, B-IV, C-I, D-II

(2) A-III, B-II, C-I, D-IV

(3) A-I, B-II, C-III, D-IV

(4) A-III, B-I, C-II, D-IV

69) Match the List-I with List-II

List-I		**List-II**	
A	Heart	I	Atrial natriuretic factor
B	Liver	II	erythropoietin
C	Kidney	III	Angiotensinogen
D	GIT	IV	CCK

Choose the correct answer rom the options are given below

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-I, C-III, D-IV

(3) A-III, B-II, C-I, D-IV

(4) A-I, B-III, C-II, D-IV

70) Match List-I with List-II:

List-I		**List-II**	
A	Neurohypophysis	i	Growth hormone
B	Acromegaly	ii	Adrenal cortex
C	Pineal gland	iii	Oxytocin
D	Addison's disease	iv	melatonin

Choose the correct answer from the options given below

(1) A-iii, B-i, C-ii, D-iv

(2) A-i, B-iii, C-ii, D-iv

(3) A-iii, B-i, C-iv, D-ii

(4) A-iv, B-i, C-ii, D-iii

71) Match List-I with List-II and select the correct option.

	List-I		**List-II**
(A)	Occipital bone	(I)	Sternum
(B)	Clavicle	(II)	Ribs
(C)	Manubrium	(III)	Pectoral girdle
(D)	Bicephalic	(IV)	Skull

Options :-

(1) A-I, B-II, C-III, D-IV

(2) A-IV, B-III, C-II, D-I

(3) A-III, B-IV, C-I, D-II

(4) A-IV, B-III, C-I, D-II

72) Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): Our vertebral column is formed by 26 serially arranged units called vertebra and is dorsally placed.

Reason (R): Vertebral coloumn extends from base of the skull and constitutes the main frame work of trank

Choose the correct answer from options given below:

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

73) **Assertion :-** Humans are sexually reproducing and viviparous.

Reason :- In human, during fertilization male & female gametes fuses and leads to zygote formation, which grows to embryo and ultimately new young one is delivered out.

(1) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(2) Both Assertion & Reason are False

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

74) **Assertion :-** Each testis has 250 compartments called testicular lobules.

Reason :- Each lobule contains two to four highly coiled seminiferous tubules in which sperms are produced.

(1) Assertion is True but the Reason is False.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Both Assertion & Reason are False

(4) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

75) **Statement (I) :** During cleavage there is no cytoplasmic growth.
Statement (II) : Cleavage is assented for zygote to from multicellular embryo.

(1) Statement (I) is true and statement (II) is false.

(2) Statement (I) is false and statement (II) is true.

(3) Both statement (I) and (II) are false.

(4) Both statement (I) and (II) are true.

76) **Statement-I :** In human beings by the end of nine month of pregnancy, the foetus is fully developed.
Statement-II : The human pregnancy lasts 9 months.

(1) Both statement-I and statement-II are incorrect

(2) Both statement-I and statement-II are correct

(3) Statement-I is incorrect but statement-II is correct

(4) Statement-I is correct but statement-II is incorrect

77) **Statement-I :** Corpus luteum secretes only estrogen which is essential for maintenance of endometrium.
Statement-II : Secretions of epididymis, vas deferens, seminal vesicle and prostate are essential for maturation and motility of sperms.

(1) Statement-(I) is correct & statement-(II) is incorrect.

(2) Statement-(I) is incorrect & statement-(II) is correct.

(3) Both statement-(I) & (II) are incorrect.

(4) Both statement-(I) & (II) are correct.

78) Identify the figure - A and figure - B and select the correct option ?



(A)



(B)

Option :

	Figure - A	**Figure - B**
1	LNG - 20	CuT
2.	Condom for male	Condom for female
3	Condom for female	Implants
4.	Implants	Cervical cap

(1) 1

(2) 2

(3) 3

(4) 4

79) Choose the incorrect one for given diagram of a contraceptive method. 

(1) It is implant used for contraception.

(2) In this progestogens alone or in combination with estrogen can also be used.

(3) Mode of action is similar to that of female condoms.

(4) They have much longer effective periods.

80) How many of the following processes are used in test tube baby programme
IVF, ET, GIFT, ZIFT, IUT, IUI, AI,

(1) Two

(2) Three

(3) Four

(4) Five

81) In human male, reproductive events occur :-

(1) Before puberty

(2) Before birth

(3) After puberty

(4) During birth

82) Administration of progesterone, progesterone oestrogen combination or IUDs are effective within…… hours of coitus as emergency contraceptives.

(1) 72

(2) 48

(3) 24

(4) 96

83) **Assertion (A) :** CNS is the site where the information is processed and controlled.
Reason (R) : The autonomic nervous system has two divisions which are antagonist to each other i.e. sympathetic and parasympathetic.

(1) Assertion & Reason both are correct and Reason is explanation of Assertion.

(2) Assertion & Reason both are correct but Reason is not a explanation of Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion & Reason are incorrect.

84) Given below are two statements:
Statement I : The somatic neural system relays impulses from the CNS to skeletal muscles while the autonomic neural system transmits impulses from the CNS to the involuntary organs and smooth muscles of the body.
Statement II : PNS comprises of all the nerves of the body associated with the brain only.
In the light of above Statements choose the correct answer from the options given below:

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

85) A patient has normal neural responses, but his doctor notes that certain functions in his body require continuous regulation, which cannot be achieved by neural control alone. The doctor explains that a separate system involving secretions from organs like the gastrointestinal tract, kidney and heart is also involved in coordination. Which system is the doctor referring to ?

(1) Respiratory system

(2) Muscular system

(3) Endocrine system

(4) Excretory system

86) Hormones are ___A___ chemicals which acts as ___B___ messengers and are produced in ___C___ amounts.

	A	B	C
(1)	Nutrient	Intracellular	Trace
(2)	Nutrient	Inter-cellular	Trace
(3)	Non-nutrient	Intercellular	Trace
(4)	Non-nutrient	Intercellular	large

(1) 1

(2) 2

(3) 3

(4) 4

87) **Assertion (A) :** Unmasking of the active site on actin occurs during muscle contraction
Reason: (R): During the muscles contraction myosin head binds to the unexposed active site on actin to form a cross bridge.

(1) Both **Assertion** and **Reason** are true but **Reason** is not the correct explanation of **Assertion**.

(2) **Assertion** is correct but **Reason** is incorrect.

(3) **Assertion** is incorrect but **Reason** is correct.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

88) Read following terminology of column-I and column-II and identify their correct match

Column-I		Column-II	
A	LH surge	i	Trophoblast attached to endometrium
B	Implantation	ii	Act on sertoli cells help in spermatogenesis
C	Foetal ejection reflex	iii	Chorionic villi formation
D	F.S.H.	iv	Rupture of graafian follicle
		v	Oxytocin release (Maternal pituitary)
		vi	Ovulation
		vii	Blastocyst embedded in the endometrium

(1) A-vi,ii ; B-i, vii ; C-v ; D-iii, iv

(2) A-iv ; B-i ; C-v, iii ; D-iii, iv

(3) A-iv,vi ; B-i,vii,iii ; C-v ; D-ii

(4) A-i,vi ; B-vii, iii ; C-iv,v ; D-ii

89) Identify the correct match from the column-I, II and III :-

I		II		III	
A	Progesterone	a	Corpus luteum	i	Maintain the endometrium
B	Oestrogen	b	Ovary	ii	Help in parturition
C	Relaxin	c	Theca interna	iii	Milk ejection reflex
D	Oxytocin	d	Posterior pituitary	iv	Proliferation of endometrium

(1) A-a-i, B-d-ii, C-b-iii, D-c-iv

(2) A-a-i, B-c-iv, C-b-ii, D-d-iii

(3) B-a-iv, A-b-iii, C-c-ii, D-d-i

(4) B-c-iii, C-b-ii, A-d-iv, D-a-i

90) Match the column-I and column-II and choose the correct match from given options :-

	Column-I		Column-II
(A)	Amniocentesis	(i)	Non-steroidal pill
(B)	Natural centraceptive	(ii)	Detection of certian genetic disorders
(C)	In-vitro fertilization	(iii)	Periodic abstinence
(D)	Saheli	(iv)	Test tube baby

	A	B	C	D

(1)	(iv)	(iii)	(i)	(ii)
(2)	(iv)	(ii)	(i)	(iii)
(3)	(ii)	(iv)	(i)	(iii)
(4)	(ii)	(iii)	(iv)	(i)

(1) 1

(2) 2

(3) 3

(4) 4

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	2	3	2	1	3	4	3	4	3	2	1	3	3	4	1	3	2	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	3	2	3	3	2	2	2	3	2	1	3	4	4	4	4	1	4	4	3
Q.	41	42	43	44	45															
A.	2	2	3	1	2															

CHEMISTRY

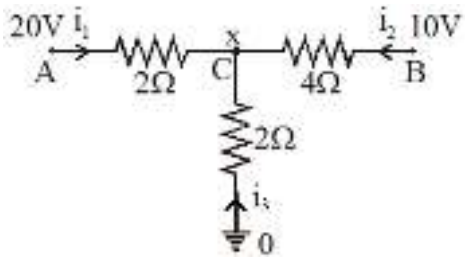
Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	4	1	2	1	3	1	2	2	2	3	4	3	2	2	2	3	2	3	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	3	2	1	4	2	3	2	3	3	4	1	2	1	2	4	2	2	3
Q.	86	87	88	89	90															
A.	4	2	2	2	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	1	3	1	1	3	2	1	1	4	3	3	3	3	2	2	4	4	3	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	1	4	4	1	1	4	3	4	2	1	3	4	1	4	1	4	3	1	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	1	3	4	3	3	1	2	4	4	1	2	3	2	4	4	3	3	3	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	1	1	3	4	2	3	3	4	3	4	2	4	1	4	2	2	2	3	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	1	2	2	3	3	2	3	2	4										

SOLUTIONS

PHYSICS



1)

Let voltage at C = x

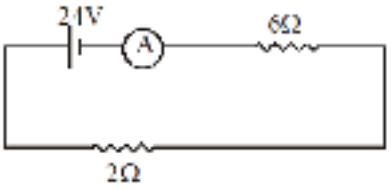
At Junction : $i_1 + i_2 + i_3 = 0$

$$\frac{20-x}{2} + \frac{10-x}{4} + \frac{(0-x)}{2} = 0$$

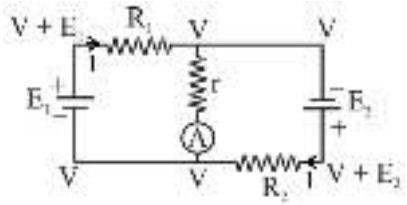
$\Rightarrow x = 10$

and i = 5 amp.

2) The circuit can be redrawn as



$$I = \frac{24}{6+2} = 3A$$



3)

Current through R_1 and R_2 will be same

$$I = \frac{E_1}{R_1} = \frac{E_2}{R_2}$$

$$\frac{E_1}{E_2} = \frac{R_1}{R_2}$$

4) $i = \frac{dQ}{dt}$ = slope of Q – t graph

= –5 (which is constant)

Amount of heat generated in time t

$H = i^2Rt$

$H \propto t$.

5) $I_g = \frac{3}{50+2950} \propto 30$, $Ig' = \frac{3}{50+R} \propto 20$

$$\Rightarrow \frac{50+R}{50+2950} = \frac{3}{2}$$

⇒ 50 + R = 4500
⇒ R = 4450Ω

6) On increasing the temperature of a conductor, resistance increases which increases the balancing length.

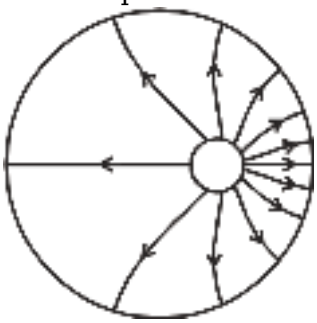
7)

mass (m) = volume × density = V(d) = A □ d

Hence $A \propto \frac{m}{\ell}$ but $R \propto \frac{\ell}{A} \propto \frac{\ell}{\frac{m}{\ell}} \propto \frac{\ell^2}{m}$

$R_1 : R_2 = \frac{(2/3)^2}{3/2} = \frac{\frac{4}{9}}{\frac{3}{2}} = 8 : 27$

8) Since both spherical surface are equipotential. The electric lines of force should be normal to both spherical surfaces. The correct choice is



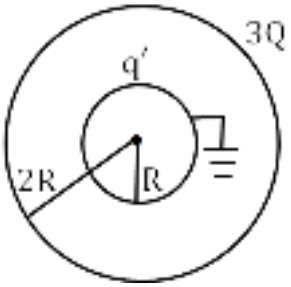
9)

Force on proton is in +x direction and path is parabolic so electric field will be uniform.

10) The net potential at the point x-along x-axis from the point charge at a point A will be :
$V_{net} = V_A + V_B$

11)

Explain Question:



If inner sphere is earthed, calculate charge (q').

Concept: Earthing of conductors.

Solution:
Potential of inner sphere given by

$$V_{\text{inner sphere}} = \frac{kq'}{R} + \frac{k(3Q)}{2R} = 0$$

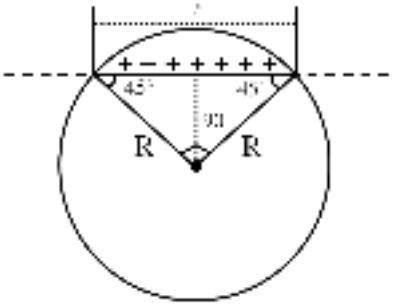
$$\frac{k}{R}\left[q' + \frac{3Q}{2}\right] = 0$$

$$q' = \frac{-3Q}{2}$$

Final Answer: Option (3)

12)

Explain the Question : We have to calculate electric flux linked with the sphere , if the rod charge density 'λ' passes through it.



Concept : This question is based on ϕ(electric flux) = $\frac{q_{inside}}{\varepsilon_0}$
Solution : from the diagram
$q_{inside} = \lambda \ell$
Calculation of □ : →

□ = 2 Rcos45° ⇒ $\lambda = 2R \times \frac{1}{\sqrt{2}} = \sqrt{2}R$

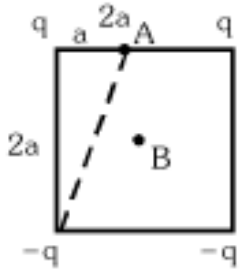
$q_{inside} = \lambda(\sqrt{2}R)$

$\phi = \frac{\lambda(\sqrt{2}R)}{\varepsilon_0}$

Final answer : Option (2)

13)

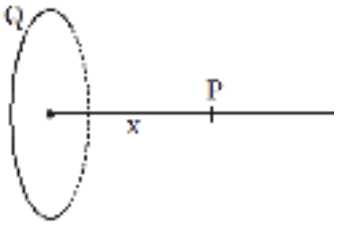
$\Delta K = W = Q(V_A - V_B)$



$$= Q\left[\left(\frac{2kq}{a} - \frac{2kq}{a\sqrt{5}}\right) - 0\right]$$

$$= \frac{2KqQ}{a}\left(1 - \frac{1}{\sqrt{5}}\right)$$

14)



$$V = \frac{kQ}{\sqrt{R^2 + x^2}} = \frac{k(3q)}{\sqrt{R^2 + 8R^2}} = \frac{kq}{R}$$

$W_E = KE_f - KE_i$

$$-(-q)(V_f - V_i) = \frac{1}{2}mv^2 - 0$$

$$q\left(\frac{kq}{R} - 0\right) = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2kq^2}{mR}} = \sqrt{\frac{q^2}{2\pi\varepsilon_0 mR}}$$

15) (i) is incorrect because '1' and '2' are replelled by '3' so '1' and '2' have charge of same nature. Hence they should repel each other. For this reason, (ii) is correct.
(iii) is correct if charge on '2' and '3' is of same nature and on '1' opposite of them.
(iv) All attraction is not possible.

16) $$\frac{\in_0 A}{d - \frac{d}{2}\left(1 - \frac{1}{k}\right)} = \frac{4 \in_0 A}{3 \quad d} \Rightarrow K = 2$$

17) $$V_C = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$

$$U_{loss} = \frac{1}{2}\frac{C_1C_2}{C_1 + C_2}(V_1 - V_2)^2$$

18)

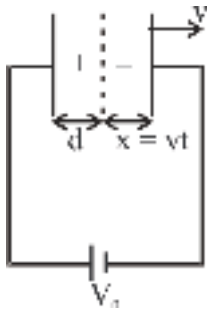
Applying KVL

$$V_A - \frac{q}{1\mu} - 10 - \frac{q}{2\mu} = V_B$$

$$\Rightarrow V_A - V_B - 10 = \frac{3q}{2\mu} \Rightarrow -20 = \frac{3q}{2\mu}$$

$$q = -\frac{40}{3}\mu C$$

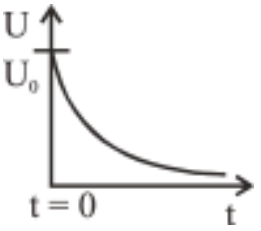
$$|q| = \frac{40}{3}\mu C$$

19) 

$U = \frac{1}{2}CV_0^2$

$U = \frac{\frac{1}{2}\varepsilon_0 AV_0^2}{(d+x)}$

$U = \frac{\varepsilon_0 AV_0^2}{2(d+vt)}$



20) $PV = nRT \Rightarrow T_2 = 16T_1$

$\Rightarrow \quad V_{2\,rms}^2 = \frac{3RT_2}{M} = 16\, V_{1\,rms}^2.$

21) $\frac{W}{Q} = \frac{nR\Delta T}{nC_p\Delta T} = \frac{32}{80} = \frac{2}{5}$

$5R = 2C_p = 2[C_v + R]$

$C_v = \frac{3R}{2}$ (Means monatomic gas)

22) $\Delta\ell = \frac{Mg\ell}{AY} = \ell\alpha\Delta T \Rightarrow M = \frac{YA\alpha\Delta T}{g}$

23) $\frac{x - LFP}{UFP - LFP} = \text{const.}$

$\frac{5-(-20)}{180-(-20)} = \frac{C-0}{100-0}$

$= \frac{25}{200} = \frac{C}{100} \Rightarrow \boxed{C = 12.5^oC}$

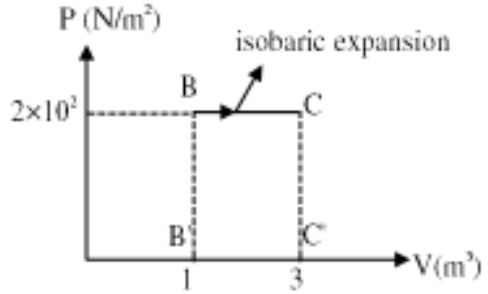
24) Rate of cooling $\propto (T^4 - T_0^4)$

$\frac{R}{R_1} = \frac{[(900)^4 - (300)^4]}{[(1200)^4 - (300)^4]}$

$$= \left(\frac{300}{300}\right)^4 \left[\frac{3^4-1^4}{4^4-1^4}\right] = \frac{80}{255}$$

$$R_1 = \frac{255R}{80}$$

25)



(diagram indicating only isobaric expansion)
work done = Area of BCC 'B'
$= 2 \times 10^2 (3 - 1)$
work done $= 4 \times 10^2$ Joule
= 400 Joule

26) $\frac{3K(100-\theta)}{L} = \frac{2K(\theta-50)}{L} + \frac{K(\theta-0)}{L}$

$300 - 3\theta = 2\theta - 100 + \theta$

$400 = 6\theta$

$\theta = \frac{200}{3}°C$

27)

$Q = \mu C \Delta T$
where μ = moles
ΔT = Change in temperature
C = Molar specific heat.
also Q = W – U
$U_A = U_B$
while $W_A = 2W_B$

28) Velocity of system after collision $v_1 = \frac{mv}{2m} = \frac{v}{2}$

let amplitude of SHM be A then $\frac{1}{2} KA^2 = \frac{1}{2} (2m) \left(\frac{v}{2}\right)^2$

$\Rightarrow A = v\sqrt{\frac{m}{2K}}$

29) The pendulums will be in phase if
$N\sqrt{5} = (N-1)\sqrt{20}$
N : no. of oscillations completed by small pendulum

no. of oscillations by shorter pendulum = N = 2

30) $mg = m\omega^2 A$

$\omega = \sqrt{\frac{g}{A}} \Rightarrow T = 2\pi\sqrt{\frac{A}{g}}$

31) $v = \omega\sqrt{A^2 - x^2}$

at $x = \frac{2A}{3}$

$v = \omega\sqrt{A^2 - \left(\frac{2A}{3}\right)^2} = \frac{\sqrt{5}A\omega}{3}$

New amplitude = A'

$v' = 3v = \sqrt{5}A\omega = \omega\sqrt{(A')^2 - \left(\frac{2A}{3}\right)^2}$

$A' = \frac{7A}{3}$

32) $U = a + bx^2$

$F = -\frac{dU}{dx} = -\frac{-[a + 2bx]}{dx}$

$F = ma = -2bx$

$\Rightarrow a = -\left(\frac{2b}{m}\right)x$ $\square$ $a = -\omega^2 x$

$\omega = 2\pi f = \left(\frac{2b}{m}\right)^{\frac{1}{2}}$

Hence frequency depends upon b and m

33)

$a_{max} = \omega^2 A \Rightarrow 40 = \omega^2 \times 10$

$\omega = 2$rad/sec $\Rightarrow x = A\sin(\omega t + \phi)$

at $t = 0$ $x = +5$

$5 = 10\sin\phi$

$\Rightarrow \sin\phi = \frac{1}{2} \Rightarrow \phi = 30°$

$x = 10\sin(2t + 30°)$

$x = 10\cos(2t - 60°)$

34) $f_1 = \frac{3v}{4L}$

$f_2 = \frac{2v}{2\left(L - \frac{L}{6}\right)} = \frac{6v}{5L}$

$\frac{f_1}{f_2} = \frac{3}{4} \times \frac{5}{6} = \frac{5}{8}$

35) $\upsilon = \sqrt{gy}$
here y = L/2

36) Frequency tone emitted = n

$$n = \frac{x}{2L}\sqrt{\frac{T}{\mu}}$$ (x = 1,2,3,4...)

If T' = 4T and L' > L

Then $$n' = \frac{x}{2L'}\sqrt{\frac{4T'}{\mu}} \approx 2n$$

Since the wire is slightly extensible therefore frequency would be slightly less than 2n

37) $$\alpha = \frac{2\pi}{\lambda} = \frac{2\pi}{0.08} = 25\pi$$

$$\beta = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$$

38)

The frequency is a characteristic of source. It is independent of the medium.

39)

Energy density of wave is given by

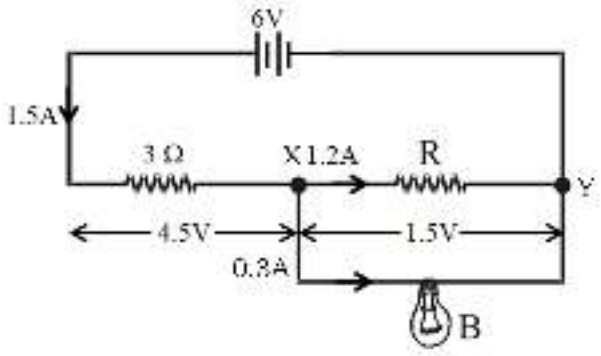
$u = 2\pi^2 n^2 \rho a^2$

Or $u \propto a^2$ (as n and ρ are constant)

$$\frac{u_1}{u_2} = \frac{a_1^2}{a_2^2} = \frac{5^2}{2^2} = \frac{25}{4}$$

40)

$$e = \frac{\ell_2 - 3\ell_1}{2}$$



41)

P = Vi

0.45 = 1.5 × i

i = 0.3 A

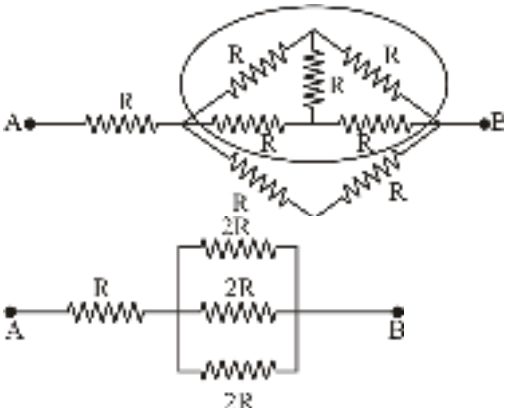
$$\frac{3}{R_{XY}} = \frac{4.5}{1.5} \Rightarrow R_{XY} = 1\,\Omega$$

42)

$$P_{maximum} = \frac{E^2}{4r} = \frac{(4)^2}{4 \times 1} = 4W$$

so power should be less than or equal to 4W

43) Balanced W.S.B.



$$R_{AB} = R + \frac{2R}{3} = \frac{5R}{3}$$

44) Volume is constant

$$R = \rho\frac{L}{A}$$

$V_1 = V_f$

$A_1L_1 = A_2L_2$

$A_1L_1 = A_2 1.001$

$$A_2 = \frac{A_1}{1.001}$$

$$R_{new} = \frac{\rho L_{new}}{A_{new}}$$

$$R_{new} = \rho\frac{1.001L}{A_{1/1.001}} = R(1.002)$$

% change in R

$$\Rightarrow \frac{1.002R - R}{R} \times 100$$

$\Rightarrow 0.2\%$

45)

Mass of copper coin

$m_c = 40\ g = 0.040\ kg, c_c = 420\ J/kg \cdot K, \Delta T_c = 75 - 25 = 50\ K$

Heat lost by copper coin:

$Q_{lost} = m_c c_c \Delta T_c = 0.040 \times 420 \times 50 = 840\ J$

Water equivalent = 20 g = 0.020 kg water (specific heat } $c_w = 4200$ J/kg.K

Heat gained by calorimeter = $0.020 \times 4200 \times (25 - T)$

Heat lost by hot body = Heat gained by cold body

$84 \times (25 - T) = 840$
$25 - T = \frac{840}{84} = 10$
$T = 25 - 10 = 15\ ^\circ C$

CHEMISTRY

46) $\frac{w_{Ag}}{E_{Ag}} = \frac{w_{Cu}}{E_{Cu}}$
$\frac{27}{\frac{108}{1}} = \frac{w}{\frac{63.3}{2}}$
$w = \frac{63.3}{2} \times \frac{27}{108}$
= 7.95 gm

47)

$E^0 = \frac{0.44 \times 2 + (-0.77) \times 1}{3} = \frac{0.88 - 0.77}{3}$
$= \frac{0.11}{3} = 0.036$

48)

$E_{cell} = E^0_{cell} - \frac{0.059}{2} \log \frac{[Cd^{+2}] \cdot pH_2}{[H^+]^2}$
$= 0.40 - 0.03 \log \frac{0.1}{(0.1)^2} \times 0.5$
$= 0.40 - 0.03 \log 5$
$= 0.40 - 0.03 \times 0.7$
$= 0.40 - 0.021$
$= 0.38$

49)

$E_{cell} = E^0_{cell} - \frac{0.059}{2} \log \frac{[Sn^{+2}]}{[Ag^+]^2}$
$[Ag^+] \uparrow$ then $E_{cell} \downarrow$

50)

$E_{cell} = E^0_{cell} - \frac{0.059}{2} \log \frac{[H^+]^2 P_{H_2}(c)}{[H^+]^2 P_{H_2}(a)}$
$= 0 - 0.03 \log \frac{P_2}{P_1}$

For E_{cell} +ve $P_1 > P_2$

51)

b, d, e are corrects.

52) Sn^{+4}/Sn^{+2} has lowest E^0_{RP} so
Sn^{+2} is strongest reductant.

53)

		At anode	At cathode
	$CuSO_4$ solution using Pt electrode	O_2	Cu
	$CuSO_4$ solution using Cu electrode	Cu	Cu
	NaCl solution using Pt electrode	Cl_2	H_2
	NaCl solution using Hg electrode	Na	Cl_2

 Option (2) is correct.

54) $[OH^-] = C\alpha$

$= 10^{-2} \cdot \frac{\Lambda_m}{\Lambda_m^\infty}$

$= 10^{-2} \times \frac{10.2}{\Lambda_m^\infty}$

$\Lambda_m^\infty(NH_4OH)$

= [130 + 320 – 110]

= 340

$= 10^{-2} \times \frac{10.2}{340} = 3 \times 10^{-4}$

pOH = 4 – 0.48 = 3.52

pH = 10.48

55)

$\wedge_m$ vs $\sqrt{C}$ straight line for S.E only so option (2) is correct.

56)

$Mg^{+2} + 2e^- \rightarrow Mg$

$Al_2^{+3} + 6e^- \rightarrow 2Al$

$n_{mg} = \frac{36}{24} = 1.5$

$n_{Al} = \frac{54}{27} = 2$

So 3F and 6F
required for Mg & Al respectively.

57)

$$\wedge_m = \frac{k \times 1000}{M}$$

$k = G \times G^*$

$G^* = 125\ m^{-1}$

$$= \frac{125}{100}\ cm^{-1}$$

$$= \frac{1}{250} \times \frac{125}{100}$$

$$= \frac{1}{200}$$

$$\wedge_m (Scm^2mol^{-1}) = \frac{\frac{1}{200} \times 1000}{1} = 5$$

58)

HCOOH and KOH
WA S.B
So phenophthalein is best indicator

59)

gm-eq $KMnO_4$ = gm-eq oxalic acid

$n \times f \times v = n \times f \times v$

+3 +4

$H_2C_2O_4 \rightarrow CO_2$

$f = |2(3 - 4)| = 2$

$0.1 \times 5 \times v = 0.1 \times 2 \times 50$

$v = 20$ ml

60)

Statement (i) & (iii) one correct.

61)

$P_{gas} = K_{H\ gas} \cdot X_{gas}$

$760\ mm\ Hg = 2 \times 10^5 mm\ Hg \cdot X_{gas}$

$$X_{gas} = \frac{760}{2} \times 10^{-5}$$

$= 380 \times 10^{-5} = 3.8 \times 10^{-3}$

62)

$\Delta T_b = i\, K_b m$

im or in ↑	ΔT_b ↑ then T_b ↑ : Concept used		
Glucose 1 % $\frac{w}{v}$	i 1	n $\frac{1}{180}$	i n $\frac{1}{180}$
Sucrose 1 % $\frac{w}{v}$	1	$\frac{1}{342}$	$\frac{1}{342}$
NaCl 1 % $\frac{w}{v}$	2	$\frac{1}{58.5}$	$\frac{1}{30}$ (Approx)
$CaCl_2$ 1% $\frac{w}{v}$	3	$\frac{1}{111}$	$\frac{1}{37}$ (Approx)

So NaCl has highest boiling point.

63)

$\frac{P_B^0 - P}{P_B^0} = x_{solute}$

$\frac{P_B^0 - 0.95P_B^0}{P_B^0} = x_{solute}$

$x_{solute} = 0.05$

$x_{solvent} = 0.95$

$\frac{n_A}{n_B} = \frac{x_A}{x_B} = \frac{5}{95}$

$\frac{w_A}{m_A} \times \frac{m_B}{w_B} = \frac{1}{19}$

$\frac{w_A}{w_B} = \frac{1}{19} \times \frac{m_A}{m_B} = \frac{1}{19} \times \frac{m_A}{0.3m_A} = \frac{1}{5.7}$

Hence $\frac{w_B}{w_A} = 5.7$

64)

$P_S = P_A^0 \cdot x_A + P_B^0 \cdot x_B$

$SP = P_A^0 \times 0.2 + 0.8 \times P_B^0$

SP = SR + SQ

65) $\pi = \frac{n}{V} \times RT \quad \Rightarrow \quad \pi = \frac{W}{M} \times \frac{RT}{V} = \frac{CRT}{M}$

(C ⇒ in g/L)

$\therefore \log\pi = \log C + \log\left(\frac{RT}{M}\right)$

Thus OA = $\log(x \times 10^{-2}) = \log\left(\frac{RT}{M}\right)$

$\Rightarrow \quad x \times 10^{-2} = \frac{RT}{M}$

$\Rightarrow x \times 10^{-2} = 0.0821 \times \frac{300}{300}$

$\Rightarrow x \times 10^{-2} = 8.21 \times 10^{-2}$

$\Rightarrow x = 8.21$

66) **Explanation:** Choose correct statement
Concept: This question is based on Collision Theory
Solution: (1) The pre-exponential factor (A) can also depends on temperature due to factors like molecular orientation and frequency to collision.
(2) Arrehenius equation:
$K = Ae^{-Ea/RT}$
As T↑, Exponential term $\left(e^{-Ea/RT}\right)\uparrow$, i.e. more molecular have enough energy to over come barrier
Final Answer: Option (4)

67) $K = \frac{1}{t} \ln \frac{100}{100 - x}$

$K = \frac{1}{100} \ln 4$ .....(1)

$K = \frac{1}{t} \ln \frac{100}{10}$ .....(2)

From eq. (1) and (2)

$\frac{1}{100} \ln 4 = \frac{1}{t} \ln 10$

$t = \frac{100}{2 \times 0.3}$

68)

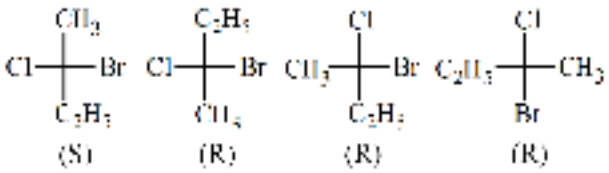
For zero order: $t_{3/4} = 1.5\ t_{1/2}$

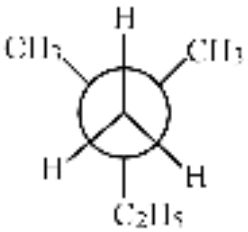
69)

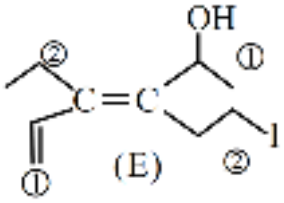
Both have different no. of C in chain.

70)

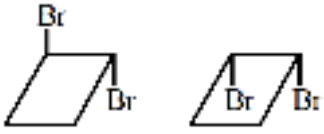
Both compounds have R configuration.

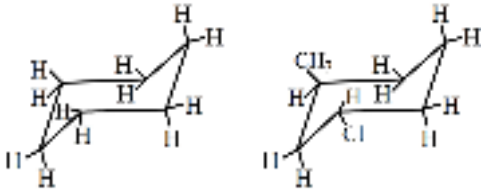
71) 

72)  2,2-Di-methyl butane

73) 

74)  having no chiral centre

75)  are cis and trans

76) 

77) **Solution:**
o-Nitrophenol: Has intramolecular hydrogen bonding → less polar → more volatile → can be distilled with steam.
p-Nitrophenol: Has intramolecular hydrogen bonding → higher boiling point → less volatile → does not distill easily with steam.
Final Answer: Option (4).

78)

Question Explanation:
The question asks to match laboratory separation techniques (List-I) with specific chemical mixtures (List-II) based on their physical and chemical properties.

Concept:
This question is based on Crystallisation, Sublimation, Simple Distillation, Chromatography, Purification of Organic Compounds.

SOLUTION :

A. (A) Crystallisation → (P, R):
Used to purify solids.

B. Sugar + water (P): sugar crystallises from solution.

C. Sugar + charcoal (R): charcoal is insoluble, sugar is recovered by filtration and crystallisation.

D. (B) Sublimation → (Q):
Separates a sublimable solid from a non-sublimable one.

E. Sugar + naphthalene (Q): naphthalene sublimes, sugar remains.

F. (C) Distillation → (S):
Separates miscible liquids with large boiling point differences.

G. Chloroform + aniline (S): Δb.p. ~123°C → simple distillation works.

H. (D) Chromatography → (T):
Separates components based on differential movement through a stationary phase.

I. Glycine + alanine (T): standard method for amino acids.

Final Answer: Option (1)

79)

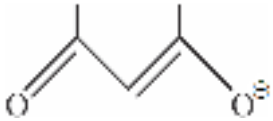
NCERT XI Pg. No. # 265

80) **Question Explanation:** Identify % of Bromine in given compound.
Concept: This question based on Carius Method
Solution:

$$\% \text{ of Br} = \frac{\text{At wt of Br}}{\text{wt of AgBr}} \times \frac{m}{w} \times 100$$

$$= \frac{80}{188} \times \frac{0.12}{0.15} \times 100 = 34.04\%$$

Final Answer: Option (1)

81)  having continuous conjugation with complete octet

82)

Stability of carbanion ∝ – M ∝ – I

83)

For aromaticity compound obey following 4 condition
(i) cyclic (ii) planar (iii) Resonance (iv) Huckle rule

84)

Resonance Structure may have diff. no. unpaired no of e^-

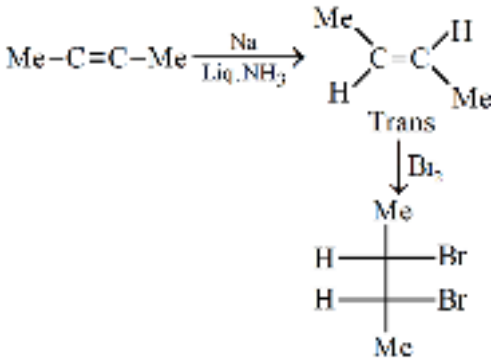
85)

Hoffmann product is formed by Anionic T.S.

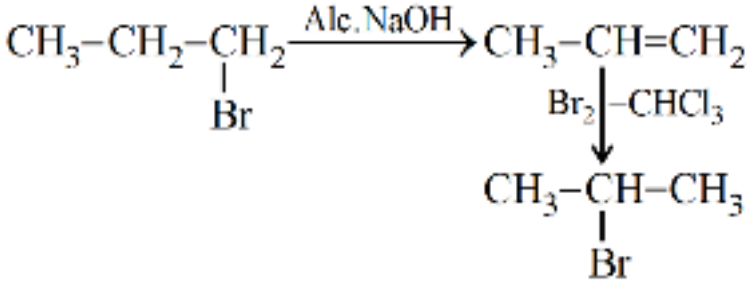
86)

E_2 is a anti elimination

87)



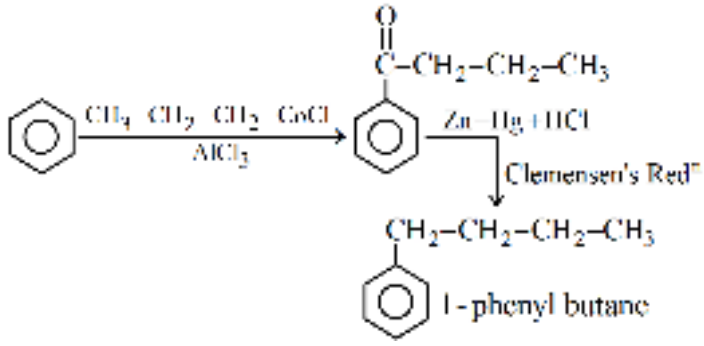
88)



89)

–I effect of Cl is more dominant than +ve mesomeric effect.
So –Cl is d +M but (o,p-directing) but deactivating group.

90)



BIOLOGY

91) NCERT XI, Pg. # 137,138

92) NCERT-XI Pg # 149,150

93)

NCERT-XI, Pg # 146

94) NCERT XI, Pg. # 141

95)

NCERT XI, Pg # 138, 139

96) NCERT XI, Pg. # 140

97) NCERT XI, Pg. # 145

98) NCERT XI, Pg. # 147

99)

NCERT XI, Pg. # 159

100) NCERT Pg. # 159

101)

NCERT XI Pg.# 161

102)

NCERT XI, Pg. # 162

103) NCERT XI, Pg. # 156

104)

NCERT XI, Pg. # 164

105)

NCERT XI, Pg. # 160

106) NCERT XI, Pg. # 171

107) NCERT XI, Pg. # 174

108)

NCERT XI, Pg. # 176

109)

NCERT XI, Pg. # 175,176,177

110)

NCERT XI, Pg. # 135

111)

NCERT XI, Pg. # 177

112)

NCERT XI, Pg. # 170

113)

NCERT XI, Pg. # 173

114) Module

115) NCERT XI, Pg. # 60

116) NCERT XI, Pg. # 64,66

117)

NCERT XI, Pg. # 64

118)

NCERT XI, Pg. # 61

119)

NCERT XI, Pg. # 62

120) NCERT XI, Pg. # 69

121) Old NCERT XI, Pg. # 95

122)

Old NCERT XI, Pg. # 86,87,88

123) NCERT XI, Pg. # 74

124) NCERT XI, Pg. # 74,75

125) NCERT XI, Pg. # 76

126) NCERT XI, Pg. # 76

127) NCERT XII, Pg. # 13

128)

NCERT XII, Pg. # 6,7

129) NCERT XII, Pg. # 13,14

130) **Explain Question :** The question asks to identify Match the fertilization event (Column I)

with its resulting product (Column II).
Concept : Double Fertilization.
Solution:

A. σ gamete + Egg cell (↑ gamete) → Zygote (Syngamy).
B. σ gamete + Secondary nucleus → Primary Endosperm Nucleus (Triple fusion).
C. The correct match is the one describing triple fusion as a process.

Final Answer: (3) σ gamete + Secondary nucleus → Triple fusion

131) NCERT XII Pg. # 18

132)

NCERT XII Pg. # 10

133) NCERT XII Pg. # 19

134) NCERT XII Pg. # 22

135) NCERT XII Pg. # 23

136) NCERT Pg.# XI- 243,242.
XII- 35,38.

137)

Nissl's granules are primarily found in the **cell body (soma)** of a neuron. In the given diagram, the cell body is labeled as **B.**
Therefore, the correct answer is **1. Only B**

138)

NCERT XI, Pg. # 232

139)

NCERT Pg. # 239

140) NCERT Pg. # 242

141) NCERT Pg. # 243

142) NCERT Pg. # 246

143) NCERT Pg. # 225

144) NCERT Pg. No. # 220

145) NCERT Pg.#221.

146) NCERT Pg. # 35

147) NCERT Pg. # 26

148) NCERT Pg.#38.

149) NCERT Pg. 38

150) NCERT Pg. 27

151) NCERT Pg. # 44

152)

NCERT XII, Pg. # 48

153) NCERT-XII, Pg. # 46, Para-3.3

154)

NCERT XI, Pg. # 234

155) NCERT XI Pg. # 230,231

156) NCERT XI Pg. 234,235

157) NCERT Pg. # 241

158) NCERT, Pg. # 242, 243, 244

159) NCERT, Pg. # 247

160) NCERT, Pg. # 241, 242, 244

161) NCERT Pg.#224-226

162) NCERT-XIth, Pg. # 225

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